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Highlights of

**Earth–Moon Transfer Trajectory Design for Solar-Sail
Spacecraft**

by

Dawei Wang, Dong Ye, Xinghong Huang, Haiyang Li, Xiao
Chen

(A paper submitted to *Advances in Space Research*)

- Develops a unified three-phase framework for fixed-endpoint solar-sail Earth–Moon transfers to L1/L2 halo orbits.
- Introduces a reverse-time solar-sailing Q-law and a solar-sail–adapted shooting formulation for energy attenuation and manifold insertion.
- Demonstrates GEO/GTO-to-halo transfers over varying area-to-mass ratios, quantifying transfer-time sensitivity and phase-wise trends.

April 1, 2026



Earth–Moon Transfer Trajectory Design for Solar-Sail Spacecraft

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Abstract

Solar-sail spacecraft provide continuous, propellantless low-thrust propulsion, but the tight coupling between sail orientation and thrust magnitude complicates the design of multi-revolution Earth–Moon transfer trajectories. This paper develops a three-phase Earth–Moon transfer design framework that combines a solar-sailing Q-Law with a forward–backward multiple-shooting scheme to construct end-to-end trajectories from Earth orbits to Earth–Moon L1/L2 halo orbits. First, the planetocentric solar-sailing Q-Law is derived based on modified equinoctial elements and reformulated as a reverse-time Q-Law. The effectiveness of solar-sailing Q-Law is demonstrated via various planetocentric transfers. Furthermore, the reverse-time Q-Law is employed in Phase 1 to drive the spacecraft from the ending point of Phase 2 to a prescribed Earth departure orbit. In Phase 2, a forward–backward multiple-shooting scheme is utilized to reduce the orbital semi-major axis while enforcing a terminal Sun phase angle close to the favorable value, thereby shortening the Phase-1 transfer. In Phase 3, a similar multiple-shooting formulation connects the preselected point on the stable invariant manifolds to the desired L1/L2 halo orbit. The three phases are finally patched to yield continuous Earth–Moon transfers. Numerical examples based on GEO and GTO departure orbits, L1 and L2 target orbits, multiple area-to-mass ratios, and both eclipse and non-eclipse conditions demonstrate the robustness and effectiveness of the proposed methodology for solar-sail mission design in the Earth–Moon system.

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Keywords: Solar sail ; Circular restricted three-body problem ; Q-Law; Earth-Moon transfer

1. Introduction

Solar-sail spacecraft provide continuous, propellantless thrust by reflecting sunlight with a large, highly reflective membrane, offering a fundamentally different mode of low-thrust propulsion compared with conventional electric or chemical systems. Their ability to generate acceleration without expending propellant makes them particularly attractive for long-duration and deep-space missions. The feasibility of solar sailing was first demonstrated in space by JAXA's IKAROS mission in 2010 [Tsuda et al., 2013]. Around the same time, the National Aeronautics and Space Administration (NASA) launched NanoSail-D2 [Johnson et al., 2011] and later launched the Near-Earth Asteroid Scout (NEA-Scout) in 2022 (which unfortunately lost contact shortly after launch) [Lockett et al., 2020]. The Planetary Society advanced the field with the LightSail program, launching LightSail-1 in 2015 [Ride-

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noure et al., 2016] and LightSail-2 in 2019 [Spencer et al., 2021]. Looking ahead, JAXA has proposed OKEANOS, a large-scale solar-power-sail mission intended for sample-return exploration of Jupiter's Trojan asteroids [Takao et al., 2023]. Reflecting the growing interest in this propulsion technology, the 13th Global Trajectory Optimisation Competition (GTOC) centered on trajectory design for solar-sail spacecraft operating within a Solar-System-like environment [Whiffen et al., 2025].

In recent years, extensive research has focused on low-thrust spacecraft performing near-Earth orbital transfers under two-body or approximately two-body conditions. Such transfers typically require dozens or even hundreds of revolutions around Earth, which makes trajectory design particularly challenging. Direct optimization methods [Howell et al., 2018; Leomanni et al., 2021; Oshima, 2023] become computationally demanding because they require a large number of discretization points to capture these long-duration, multi-revolution trajectories. Conversely, indirect optimization methods [Haberkorn et al., 2004; Fu et al., 2025; Li et al., 2025] often exhibit strong sensitivity to costate initial conditions, making it difficult to obtain convergent solutions. To address these difficulties, Ilgen [Ilgen, 1994] introduced a Lyapunov-based feedback control approach, in which a Lyapunov function is constructed to quantify the difference between the current and target orbits, and the thrust-control law is designed to reduce this function as rapidly as possible. Although the resulting transfers are generally suboptimal, the analytical structure enables fast computation [Narayanaswamy & Damaren, 2023]. Building on this idea, Petropoulos [Petropoulos, 2005, 2003] refined the method and formalized it as the Q-Law, in which the Lyapunov function is termed the "proximity quotient." Petropoulos [Petropoulos, 2004] also analyzed how the effectiveness of reducing the Q-function depends on the orbital state and introduced a coasting mechanism for conditions where thrusting yields limited benefit. When applied to solar-sail propulsion, the problem becomes more complex because the thrust direction is restricted by the Sun-sail geometry, reducing maneuvering flexibility relative to conventional low-thrust engines. Macdonald [Macdonald & McInnes, 2005b] developed an analytical solar-sail control law based on Keplerian elements, enabling orbit-transfer design with blended elementwise control laws. Niccolai et al. [Niccolai et al., 2021] extended low-thrust Q-Law concepts to solar-sail interplanetary transfers. Oguri [Oguri et al., 2023] later formulated a Keplerian-element-based Q-Law tailored specifically for many-revolution, planetocentric solar-sail transfers. More recently, Yin [Yin & Damaren, 2025] proposed a solar-sail control approach that exploits the geometric projection between the ideal low-thrust Q-Law acceleration direction and the achievable solar-sail acceleration.

Research on solar-sail dynamics in multi-body gravitational environments generally falls into two broad categories. The first concerns the analysis and construction of artificial dynamical structures enabled or enhanced by solar radiation pressure, such as dynamical substitutes of the equilibrium points [Gao et al., 2019, 2022], synodic resonant libration-point orbits [Heiligers et al., 2016, 2015; Gao et al., 2022, 2025; Chujo & Takao, 2022], and specialized orbits for continuous polar observation [Gómez Losada & Heiligers, 2021]. These studies highlight the rich set of non-Keplerian trajectories attainable through sustained SRP force and may serve as nominal references for further station-keeping control [Gao et al., 2023; Chujo, 2025]. The second category focuses on designing transfer trajectories for prospective exploration missions. For example, Gong et al. [Gong et al., 2010] designed solar-sail transfers from Earth to collinear equilibrium points within the Sun-Earth Hill frame. Ariadna et al. [Farrés, 2017; Farrés et al., 2019] constructed transfers from the Sun-Earth L1/L2 regions to the L4/L5 points by exploiting the invariant manifolds associated with these equilibria in the Sun-Earth restricted three-body problem. More recently, Chujo et al. [Chujo et al., 2023] developed a trajectory-patching approach in the Sun-Earth-Moon bicircular four-body problem to link Earth-Moon NRHO orbits with Sun-Earth L1/L2 halo orbits under solar-sail propulsion.

Nomenclature

A	solar sail area, m ²	α	solar sail cone angle
AU	astronomic unit, km	β	solar sail pitch angle
DU	distance unit in CRTBP	$\mathbf{r}$	position vector
TU	time unit in CRTBP	μ	mass parameter of CRTBP
$\mathbf{a}_{\text{SRP}}$	solar-sailing acceleration, m/s ²	μ_{E}	Earth gravity constant
f_{SRP}	solar radiation pressure force, N	μ_{M}	lunar gravity constant
ω_{S}	angular velocity of Sun	μ_{S}	solar gravity constant
S_0	solar constant, kg/s ³	ω	argument of periapsis
c	sunlight velocity in space, km/s	Ω	right ascension of the ascending node
d_{A}	Sun-sail distance in astronomic unit	p	semilatus rectum
d	dimensional Sun-sail distance, km	f	modified equinoctial element
C_1, C_2, C_3	sail optical coefficient	g	modified equinoctial element
m	mass of solar-sail spacecraft, kg	h	modified equinoctial element
σ	area-to-mass ratio, m ² /kg	k	modified equinoctial element
$\hat{\mathbf{n}}$	sail-plane normal vector	L	true longitude
$\hat{\mathbf{s}}$	sunlight unit vector	θ	phase angle
$\mathbf{u}$	vector of control variables	ω_{M}	lunar orbital angular velocity
$I\{\hat{\mathbf{i}}_x, \hat{\mathbf{i}}_y, \hat{\mathbf{i}}_z\}$	ECI frame	$S\{\hat{\mathbf{s}}_r, \hat{\mathbf{s}}_t, \hat{\mathbf{s}}_h\}$	solar sail body-fixed frame
$P\{\hat{\mathbf{p}}_r, \hat{\mathbf{p}}_t, \hat{\mathbf{p}}_h\}$	perifocal frame	$I\{\hat{\mathbf{l}}_r, \hat{\mathbf{l}}_t, \hat{\mathbf{l}}_h\}$	LVLH frame
$R\{\hat{\mathbf{r}}_x, \hat{\mathbf{r}}_y, \hat{\mathbf{r}}_z\}$	Earth-Moon barycentric rotating frame		
<i>Superscripts</i>			
I	LVLH frame	I	ECI frame
S	solar sail body-fixed frame	P	perifocal frame
R	Earth-Moon barycentric rotating frame	P1	Phase 1
P2	Phase 2	P3	Phase 3
f	forward time	b	backward time
T	matrix transpose		
<i>Subscripts</i>			
i	initial	f	final
S	Sun	E	Earth
M	Moon	t	target
d	departure	α	orbital element
sc	spacecraft	min	minimal
SRP	solar radiation pressure		

Although solar-sail trajectory design in multi-body dynamical systems has been widely explored, a complete methodology for constructing end-to-end Earth–Moon transfer trajectories for solar-sail spacecraft remains lacking. This paper introduces a three-phase trajectory design framework that enables seamless transfers from Earth departure orbits to Earth–Moon L1/L2 halo orbits with solar-sailing. A solar-sailing Q-Law is first derived based on modified equinoctial elements (MEEs) and reformulated as a reverse-time Q-Law to guide the spacecraft from an intermediate state back to a prescribed Earth departure orbit in Phase 1. A point on the stable invariant manifold of the target halo orbit is then selected a priori and used as the common initial state for Phases 2 and 3. A forward–backward multiple-shooting strategy is employed to reduce orbital semi-major axis in Phase 2. Furthermore, a similar multiple-shooting scheme is utilized to connect the preselected manifold point to the target halo orbit. Finally, the three phases are patched to produce a continuous Earth–Moon (E–M) transfer trajectory. Although developed for solar-sail propulsion, the proposed three-phase design framework can be extended to other continuous low-thrust systems, such as solar electric propulsion (SEP) or electric solar-wind sails, with proper modifications to the Phase-1 guidance formulation.

The remainder of this paper is organized as follows. Section 2 presents the dynamical models employed in this study. Section 3 develops the solar-sailing Q-Law in the MEE framework and introduces its adaptation into a reverse-time form for departure-orbit regression, together with verification of its effectiveness and convergence properties. Section 4 details the complete Earth–Moon transfer design methodology. Section 5 presents numerical simulations for transfers from GEO and GTO to Earth–Moon L1/L2 halo orbits under various area-to-mass ratios. Finally, Section 6 summarizes the conclusions of this study.

2. Preliminaries

2.1. Solar Radiation Pressure

The solar radiation pressure (SRP) force acting on the sail is given by [Oguri et al., 2022]

$$\mathbf{f}_{\text{SRP}} = \frac{S_0}{c d_A^2} A (\hat{\mathbf{n}} \cdot \hat{\mathbf{s}}) [(C_1(\hat{\mathbf{n}} \cdot \hat{\mathbf{s}}) + C_2)\hat{\mathbf{n}} + C_3\hat{\mathbf{s}}] \quad (1)$$

where $\hat{\mathbf{n}}$ and $\hat{\mathbf{s}}$ denote the unit normal vector of the sail plane and the unit vector pointing from the Sun to the spacecraft, respectively. The parameters S_0 , c , d_A , and A represent the solar constant (1366 kg/s^3), the speed of light in vacuum ($3 \times 10^5 \text{ km, s}^{-1}$), the sail–Sun distance normalized by the astronomical unit ($d_A = d/\text{AU}$, $\text{AU} = 1.496 \times 10^8 \text{ km}$, with d the dimensional distance), and the sail area. The coefficients C_1 , C_2 , and C_3 are determined by the optical properties of the sail material. Let m denote the spacecraft mass and define the area-to-mass ratio as $\sigma = A/m$. The corresponding SRP acceleration acting on the spacecraft is then

$$\mathbf{a}_{\text{SRP}} = \frac{\mathbf{f}_{\text{SRP}}}{m} = \frac{S_0}{c d_A^2} \sigma (\hat{\mathbf{n}} \cdot \hat{\mathbf{s}}) [(C_1(\hat{\mathbf{n}} \cdot \hat{\mathbf{s}}) + C_2)\hat{\mathbf{n}} + C_3\hat{\mathbf{s}}] \quad (2)$$

2.2. Equations of Motion in Earth-centered Inertial Frame

The relationship between the MEEs ($p[\text{km}]$ semilatus rectum, $f, g, h, k, L[\text{rad}]$ true longitude) and classical elements ($a[\text{km}]$ semi-major axis, e eccentricity, $i[\text{rad}]$ inclination, $\Omega[\text{rad}]$ right ascension of the ascending node, $\omega[\text{rad}]$ argument of periapsis, $\nu[\text{rad}]$ true anomaly) is given by

$$p = a(1 - e^2) \quad (3)$$

$$f = e \cos(\Omega + \omega) \quad (4)$$

$$g = e \sin(\Omega + \omega) \quad (5)$$

$$h = \tan(i/2) \cos(\Omega) \quad (6)$$

$$k = \tan(i/2) \sin(\Omega) \quad (7)$$

$$L = \Omega + \omega + \nu \quad (8)$$

The dynamics modeled by the modified equinoctial elements in the Earth-centered inertial frame (ECI) are as follows:

$$\dot{\mathbf{x}} = \mathbf{A}(\mathbf{x})\mathbf{F}^l + \mathbf{b} \quad (9)$$

$$\mathbf{A} = \frac{\sqrt{p/\mu_E}}{q} \begin{bmatrix} 0 & 2p & 0 \\ q \sin(L) & (q+1) \cos(L) + f & -g(h \sin(L) - k \cos(L)) \\ -q \cos(L) & (q+1) \sin(L) + g & f(h \sin(L) - k \cos(L)) \\ 0 & 0 & s \cos(L)/2 \\ 0 & 0 & s \sin(L)/2 \\ 0 & 0 & h \sin(L) - k \cos(L) \end{bmatrix} \quad (10)$$

$$\mathbf{b} = [0 \ 0 \ 0 \ 0 \ 0 \ \sqrt{\mu_E p} (q/p)^2]^T \quad (11)$$

$$s = 1 + h^2 + k^2 \quad (12)$$

$$q = 1 + f \cos(L) + g \sin(L) \quad (13)$$

where μ_E is the gravitational constant of Earth, $\mathbf{F}^l$ denotes the total perturbing acceleration expressed in the local-vertical–local-horizontal (LVLH) frame, the subscript 'l' indicates LVLH frame $\mathcal{I}\{\hat{\mathbf{i}}_r, \hat{\mathbf{i}}_t, \hat{\mathbf{i}}_h\}$. $\mathbf{F}^l$ contains two contributions: the lunar gravitational acceleration Δ_M^l and the SRP acceleration $\mathbf{a}_{\text{SRP}}^l$.

Fig. 1 illustrates the ECI frame, the sail body-fixed frame, the perifocal frame, and the LVLH frame. The ECI frame $\mathcal{I}\{\hat{\mathbf{i}}_x, \hat{\mathbf{i}}_y, \hat{\mathbf{i}}_z\}$ is centered at Earth, with $\hat{\mathbf{i}}_x$ pointing toward the mean vernal equinox, $\hat{\mathbf{i}}_z$ normal to the ecliptic plane toward the north ecliptic pole, and $\hat{\mathbf{i}}_y$ completing the right-hand system. The solar sail body-fixed frame $\mathcal{S}\{\hat{\mathbf{s}}_r, \hat{\mathbf{s}}_t, \hat{\mathbf{s}}_h\}$ is centered on the spacecraft, where $\hat{\mathbf{s}}_r = \hat{\mathbf{s}}$ is aligned with the sunlight direction, $\hat{\mathbf{s}}_t = \hat{\mathbf{i}}_z \times \hat{\mathbf{s}}_r / \|\hat{\mathbf{i}}_z \times \hat{\mathbf{s}}_r\|$, and $\hat{\mathbf{s}}_h$ completes the right-hand system. The perifocal frame $\mathcal{P}\{\hat{\mathbf{p}}_r, \hat{\mathbf{p}}_t, \hat{\mathbf{p}}_h\}$ is centered at Earth, with $\hat{\mathbf{p}}_r$ pointing toward periapsis, $\hat{\mathbf{p}}_h$ aligned with the orbital angular momentum vector, and $\hat{\mathbf{p}}_t$ completing the right-hand system. Finally, LVLH frame $\mathcal{I}\{\hat{\mathbf{i}}_r, \hat{\mathbf{i}}_t, \hat{\mathbf{i}}_h\}$ is centered on the spacecraft, with $\hat{\mathbf{i}}_r$ along the Earth–spacecraft line, $\hat{\mathbf{i}}_h$ along the spacecraft's orbital angular momentum vector, $\hat{\mathbf{i}}_t$ completing the right-hand system.

The lunar gravitational acceleration in ECI frame is expressed as

$$\Delta_M = -\mu_M \left(\frac{\mathbf{r}_{M-\text{sc}}}{|\mathbf{r}_{M-\text{sc}}|^3} - \frac{\mathbf{r}_M}{|\mathbf{r}_M|^3} \right) \quad (14)$$

where μ_M is the gravitational parameter of the Moon. The vectors $\mathbf{r}_{M-\text{sc}}$ and $\mathbf{r}_M$ denote the position of the spacecraft relative to the Moon and the position of the Moon relative to Earth, respectively. The Moon is assumed to move along a circular orbit around the Earth with a radius of $d_{M-E} = 384400$ km in $\hat{\mathbf{i}}_x - \hat{\mathbf{i}}_y$ plane. The position of the Moon at time t is calculated as

$$\mathbf{r}_M = d_{M-E} [\cos(\theta_{M0} + \omega_M t) \ \sin(\theta_{M0} + \omega_M t) \ 0]^T \quad (15)$$

where $\omega_M = \sqrt{(\mu_M + \mu_E)/d_{M-E}^3}$ is the orbital angular velocity of Moon, θ_{M0} is the initial phase of Moon at $t = 0$. The lunar phase angle is defined as the counterclockwise angle measured from $\hat{\mathbf{i}}_x$ to the Earth–Moon line.

The solar-sail attitude is parameterized in the body-fixed frame, as illustrated in Fig. 2. The cone angle α is defined as the angle between the sail normal $\hat{\mathbf{n}}$ and the sunlight direction $\hat{\mathbf{s}}_r$. The pitch angle β is measured counterclockwise from $\hat{\mathbf{s}}_r$ to the projection of $\hat{\mathbf{n}}$ onto the $\hat{\mathbf{s}}_r - \hat{\mathbf{s}}_h$ plane. The Sun is modeled as moving on a circular orbit around the Earth in the $\hat{\mathbf{i}}_x - \hat{\mathbf{i}}_y$ plane. In addition, the incident sunlight is assumed to be parallel throughout the Earth–Moon system; thus, at a given time t , the unit vector $\hat{\mathbf{s}}$ is taken to be identical for all spacecraft positions. Accordingly,

$$\hat{\mathbf{s}} = -[\cos(\theta_{S0} + \omega_S t) \ \sin(\theta_{S0} + \omega_S t) \ 0]^T \quad (16)$$

where $\omega_S = \sqrt{(\mu_M + \mu_E + \mu_S)/d_{S-E}^3}$ is the angular velocity of Sun, $d_{S-E} = \text{AU} = 1.496 \times 10^8 \text{ km}$ is the Sun-to-Earth distance, θ_{S0} is Sun phase angle at $t = 0$. The Sun phase angle in the ECI frame is defined as the counterclockwise angle from $\hat{i}_x$ to the direction $-\hat{s}$. Finally, the spacecraft–Sun distance is assumed to be constant across the Earth–Moon system, so that $d_A \equiv 1$. These assumptions are introduced to maintain consistency with the solar-sail-augmented circular restricted three-body model introduced in the subsequent section.

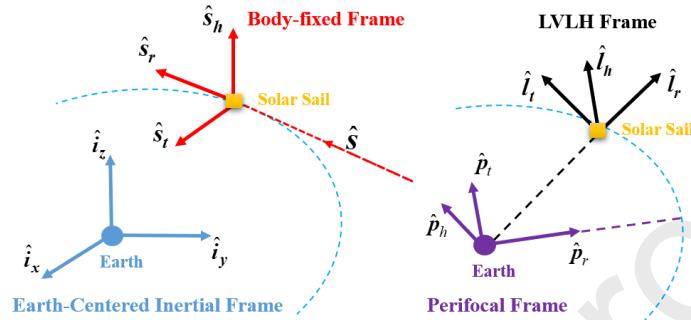


Fig. 1: Illustration of coordinate frames.

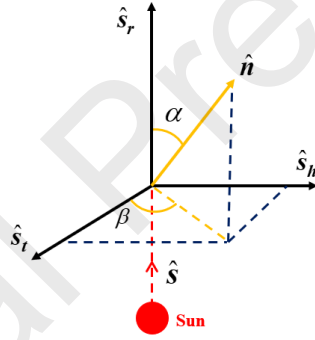


Fig. 2: Solar sail attitude angles in body-fixed frame.

2.3. Eclipse Geometry

Under the parallel-sunlight assumption adopted for the Earth–Moon space, the Earth’s shadow is modeled as an infinite cylinder aligned with the Sun–line direction, as illustrated in Fig. 3. Let $\mathbf{r}_{E-sc}$ denote the spacecraft position vector relative to the Earth’s center. The spacecraft is considered to be in eclipse only when the following two conditions are simultaneously satisfied:

$$\mathbf{r}_{E-sc}^T \hat{s} > 0 \quad (17)$$

$$\|\mathbf{r}_{E-sc} - (\mathbf{r}_{E-sc}^T \hat{s}) \cdot \hat{s}\| < 6378 \text{ km} \quad (18)$$

Eq. (17) ensures that the spacecraft lies on the anti-sunward side of the Earth. Eq. (18) requires that the perpendicular distance from the spacecraft to the Sun–Earth axis be smaller than the Earth’s radius, thereby placing the spacecraft inside the cylindrical shadow. When eclipse effects are incorporated into the dynamical model, the SRP acceleration is scaled by the shadow function ψ , defined

$$\psi = \begin{cases} 0, & \text{if Eqs. (17)–(18) are satisfied,} \\ 1, & \text{otherwise.} \end{cases} \quad (19)$$

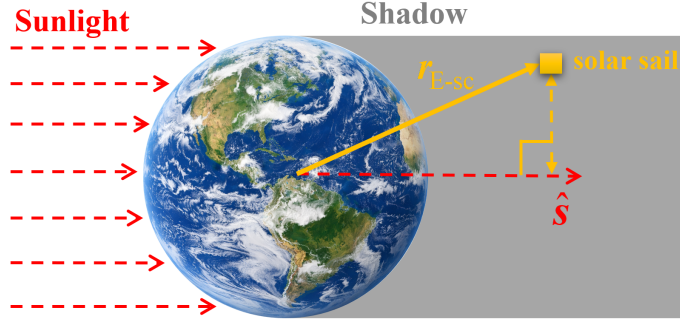


Fig. 3: Eclipse Geometry

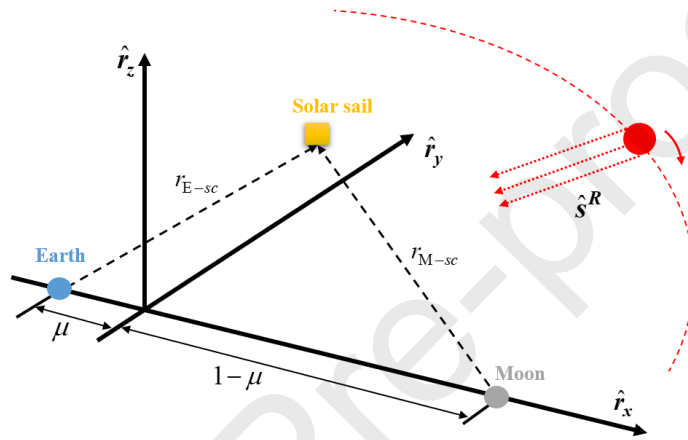


Fig. 4: Earth-Moon barycentric rotating frame.

2.4. Equations of Motion in Barycentric Rotating Frame

In the circular restricted three-body problem (CR3BP), the Earth and Moon are modeled as moving on circular orbits around their barycenter. To simplify the spacecraft equations of motion, nondimensional units are adopted. The distance unit is defined as the distance between Earth and Moon: $DU = 384400$ km, and the mass parameter is $\mu = \mu_M/(\mu_E + \mu_M)$. The time unit (TU) is defined as $TU = \sqrt{DU^3/(\mu_E + \mu_M)}$. As illustrated in Fig.4, the Earth-Moon barycentric rotating frame $R\{\hat{r}_x, \hat{r}_y, \hat{r}_z\}$ is defined to have the origin at Earth-Moon barycenter, $\hat{r}_x$ along Earth-to-Moon vector, $\hat{r}_z$ along the orbital angular momentum vector of Moon, and $\hat{r}_y$ completing the right-hand system. Thus, the Earth is fixed at $[-\mu, 0, 0]^T$, and the Moon is fixed at $[1-\mu, 0, 0]^T$. The spacecraft equations of motion in the rotating frame are

$$\begin{cases} \ddot{x} - 2\dot{y} = \frac{\partial U}{\partial x} + \mathbf{a}_{SRP,x}^R \\ \ddot{y} + 2\dot{x} = \frac{\partial U}{\partial y} + \mathbf{a}_{SRP,y}^R \\ \ddot{z} = \frac{\partial U}{\partial z} + \mathbf{a}_{SRP,z}^R \end{cases} \quad (20)$$

where $\mathbf{a}_{SRP}^R = [\mathbf{a}_{SRP,x}^R, \mathbf{a}_{SRP,y}^R, \mathbf{a}_{SRP,z}^R]^T$ represents the SRP acceleration vector expressed in $R\{\hat{r}_x, \hat{r}_y, \hat{r}_z\}$, and U is the pseudo-potential function defined as

$$\begin{aligned} U &= \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_{M-sc}} + \frac{\mu}{r_{E-sc}} + \frac{\mu(1-\mu)}{2}, \\ r_{E-sc} &= \sqrt{(x+\mu)^2 + y^2 + z^2}, \\ r_{M-sc} &= \sqrt{(x-1+\mu)^2 + y^2 + z^2} \end{aligned} \quad (21)$$

where r_{E-sc} and r_{M-sc} represent the distance from the spacecraft to the Earth and Moon respectively. In CR3BP, the Jacobi constant is defined as $J = 2U - (\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$. This constant establishes a relationship between the spacecraft's velocity and position, with higher Jacobi constants indicating lower energy levels. The modeling assumptions introduced in Sec. 2.2 are retained, namely that the sunlight direction is uniform throughout the Earth–Moon system and that the normalized Sun distance satisfies $d_A \equiv 1$. Accordingly, the sunlight direction expressed in the rotating frame $\hat{s}^R$ is written as

$$\hat{s}^R = -[\cos(\theta'_{S0} + \omega_S t), \sin(\theta'_{S0} + \omega_S t), 0]^T \quad (22)$$

where $\theta'_{S0} = \theta_{S0} - \theta_{M0} = \theta_{S0}$ is the phase angle of sun at $t = 0$ in CRTBP, ω_S is the angular velocity of Sun in CRTBP: $\omega_S = \sqrt{(1 + \bar{\mu}_S)/\bar{d}_S^3} - 1$, $\bar{\mu}_S$ is the nondimensional gravitational parameter for the sun: $\bar{\mu}_S = \mu_S/(\mu_E + \mu_M)$, $\bar{d}_S$ is the nondimensional Earth–Sun distance: $\bar{d}_S = d_{S-E}/DU$. The Sun phase angle in the rotating frame is measured counterclockwise from $\hat{r}_x$ to the direction of $-\hat{s}^R$. Given the initial Sun phase angle θ'_{S0} and the nondimensional time t in CRTBP, the corresponding Sun phase angle in the ECI frame is $\theta_S = \theta_{M0} + \theta'_{S0} + (\omega_S + 1)t$. Similarly, the corresponding moon phase angle in the ECI frame is $\theta_M = \theta_{M0} + t$. Moreover, the solar-sail body-fixed frame $S^R\{\hat{s}_r^R, \hat{s}_t^R, \hat{s}_h^R\}$ in CRTBP, is defined with its origin at the spacecraft, $\hat{s}_r^R = \hat{s}^R$, $\hat{s}_t^R = \hat{r}_z \times \hat{s}_r^R / |\hat{r}_z \times \hat{s}_r^R|$, $\hat{s}_h^R$ completing the right-hand system. Recall that the Moon is assumed to move within $\hat{i}_x - \hat{i}_y$ plane, so if $\hat{r}_z$ is converted into the ECI frame it equals $\hat{i}_z$. Therefore, the definitions of the two solar-sail body-fixed frames, $S\{\hat{s}_r, \hat{s}_t, \hat{s}_h\}$ and $S^R\{\hat{s}_r^R, \hat{s}_t^R, \hat{s}_h^R\}$, are consistent so that the attitude angles of the solar sail defined in Fig.2 remain applicable in the CR3BP. Finally, state-vector transformations between the ECI frame and the Earth–Moon barycentric rotating frame are required in the subsequent transfer design procedure. A concise description of the transformation procedure is provided in the Appendix, while the detailed derivation can be found in [Spreen, 2021].

3. Solar sailing Q-Law based on MEEs

This section derives a solar-sailing Q-Law formulation in modified equinoctial elements (MEEs) and subsequently reformulates it as a reverse-time Q-Law for use in the Earth–Moon transfer design developed in the following sections. A comparison with recent work is also provided to demonstrate the effectiveness of the proposed approach.

3.1. Proximity Quotient Function

A candidate Lyapunov function, referred to as the proximity quotient Q , is constructed to quantify the deviation between the spacecraft's instantaneous orbit and the prescribed target orbit:

$$Q = (1 + P) \sum_{\alpha} W_{\alpha} S_{\alpha} \left(\frac{\alpha - \alpha_t}{\dot{\alpha}_{mm}} \right)^2 \quad (23)$$

where $\alpha \in \{p, f, g, h, k\}$ denotes the set of slowly varying MEEs; α_t is the target orbit elements; $\dot{\alpha}_{mm}$ is the maximum attainable rate of change of α ; the coefficients W_{α} are element-wise weights; P is an exponential barrier penalty function; S_{α} is a scaling function. The functions P and S_{α} are defined as

$$P(\mathbf{x}_m) = \exp \left[k_p \left(1 - \frac{r_p}{r_{p,\min}} \right) \right] \quad (24)$$

$$S_{\alpha}(\mathbf{x}_m) = \begin{cases} \left[1 + \left(\frac{\alpha - \alpha_t}{m_{scl} \alpha_t} \right)^{n_{scl}} \right]^{1/r_{scl}}, & \alpha = p, \\ 1, & \alpha \in \{f, g, h, k\} \end{cases} \quad (25)$$

where $\mathbf{x}_m = [p, f, g, h, k]^T$, r_p is the periapsis radius of current orbit; $r_{p,\min}$ is the permissible minimum periapsis radius; $k_p = 1 \times 10^3$ is a constant for the penalty function; $m_{\text{scl}} = 3$, $n_{\text{scl}} = 4$, $r_{\text{scl}} = 2$ are the constants for scaling function.

The extreme attainable rate of change for each element, $\dot{\alpha}_{\text{mm}}$ is defined to capture the most “favorable” (in magnitude and sign) evolution toward the target. Specifically, if $\alpha < \alpha_t$, $\dot{\alpha}_{\text{mm}}$ is the maximum value of $\dot{\alpha}$ over α, β and L , and when $\alpha > \alpha_t$, $\dot{\alpha}_{\text{mm}}$ is the minimum value of $\dot{\alpha}$ over the same variables. Therefore, $\dot{\alpha}_{\text{mm}}$ is expressed as

$$\dot{\alpha}_{\text{mm}}(\mathbf{x}_m) = \dot{\alpha}(\mathbf{x}_m, L_{\alpha}^*, \alpha_{\alpha}^*, \beta_{\alpha}^*) \quad (26)$$

$$\{L_{\alpha}^*, \alpha_{\alpha}^*, \beta_{\alpha}^*\} = \arg \min_{\alpha, \beta, L} [\varepsilon_{\alpha} \dot{\alpha}(\mathbf{x}, \alpha, \beta)] \quad (27)$$

where $\mathbf{x} = [\mathbf{x}_m^T, L]^T$, $\varepsilon_{\alpha} = \text{sign}(\alpha - \alpha_t)$. Hence, $\dot{\alpha}_{\text{mm}}$ is obtained by solving the optimization problem (27).

3.2. Evaluating the Extreme Attainable Element Rates

This subsection presents an efficient approximation for computing the extreme attainable element rate $\dot{\alpha}_{\text{mm}}$. The optimization problem (27) is decomposed into two subproblems:

$$\{\alpha_{\alpha}^*(\mathbf{x}), \beta_{\alpha}^*(\mathbf{x})\} = \arg \min_{\alpha, \beta} [\varepsilon_{\alpha} \dot{\alpha}(\mathbf{x}, \alpha, \beta)] = \arg \min_{\alpha, \beta} [\varepsilon_{\alpha} (\mathbf{A}_{\alpha}(\mathbf{x}) \mathbf{a}_{\text{SRP}}^l(\alpha, \beta) + \mathbf{b}_{\alpha}(\mathbf{x}))] = \arg \min_{\alpha, \beta} [\varepsilon_{\alpha} \mathbf{A}_{\alpha}(\mathbf{x}) \mathbf{a}_{\text{SRP}}^l(\mathbf{x}, \alpha, \beta)] \quad (28)$$

$$L_{\alpha}^*(\mathbf{x}_m) = \arg \min_L [\dot{\alpha}_m(\mathbf{x}_m, L)] = \arg \min_L [\varepsilon_{\alpha} \mathbf{A}_{\alpha}(\mathbf{x}_m, L) \mathbf{a}_{\text{SRP}}^l(\mathbf{x}, \alpha_{\alpha}^*, \beta_{\alpha}^*)] \quad (29)$$

where $\mathbf{A}_{\alpha}$ is the row vector of $\mathbf{A}$ in Eq.(10) associated with α (e.g. $A_p = (\sqrt{p/\mu_E}/q)[0, 2p, 0]$). Hence, problem (28) is solved first and the intermediate rate $\dot{\alpha}_m$ is obtained by substituting $\alpha_{\alpha}^*, \beta_{\alpha}^*$ into $\varepsilon_{\alpha} \dot{\alpha}$. Next, problem (29) is solved and $\dot{\alpha}_{\text{mm}}$ is evaluated by substituting L_{α}^* into $\dot{\alpha}_m$. In general, the solution of problem (28) depends on L , and obtaining closed-form expressions for $\alpha_{\alpha}^*(L)$ and $\beta_{\alpha}^*(L)$ is difficult. To avoid solving the fully coupled problem, we first compute an approximate minimizer of (29), denoted by $\tilde{L}_{\alpha}^*$. Using $\tilde{L}_{\alpha}^*$, we then obtain approximate attitude solutions $\{\tilde{\alpha}_{\alpha}^*, \tilde{\beta}_{\alpha}^*\}$ from problem (28). Finally, an approximation of $\dot{\alpha}_{\text{mm}}$ is computed using $\tilde{\alpha}_{\alpha}^*, \tilde{\beta}_{\alpha}^*$, and $\tilde{L}_{\alpha}^*$.

First, an approximation $\tilde{L}_{\alpha}^*$ is obtained by removing the explicit dependence on the attitude angles in (29). To this end, we assume that the SRP acceleration direction is aligned with the Sun-to-sail line (i.e., along $\hat{\mathbf{s}}$) and that its magnitude is constant. Under these assumptions, problem (29) reduces to

$$\tilde{L}_{\alpha}^*(\mathbf{x}_m) = \arg \min_L [\varepsilon_{\alpha} \mathbf{A}_{\alpha}(\mathbf{x}_m, L) \hat{\mathbf{s}}^l] \quad (30)$$

The Sun-to-sail unit vector expressed in the LVLH frame, $\hat{\mathbf{s}}^l$, is obtained through the coordinate transformations

$$\hat{\mathbf{s}}^l = \mathbf{C}_P^l \mathbf{C}_I^P \hat{\mathbf{s}} = \mathbf{C}_P^l \hat{\mathbf{s}}^P \quad (31)$$

where $\mathbf{C}_I^P$ denotes the direction-cosine matrix (DCM) from the ECI frame $\mathbf{I}$ to the perifocal frame $\mathbf{P}$, and $\mathbf{C}_P^l$ denotes the DCM from $\mathbf{P}$ to the LVLH frame $\mathbf{l}$. The matrix $\mathbf{C}_P^l$ is calculated as

$$\mathbf{C}_P^l = \mathbf{R}_{\tilde{z}}(\theta) = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (32)$$

where θ is the true anomaly of the spacecraft's current orbit. From the MEE-classical element relations in Eq.(3)–(8), fixing $\mathbf{x}_m$ implies that Ω and ω are fixed. Therefore, the true longitude L and the true anomaly θ are in one-to-one correspondence: $\theta = L - \Omega - \omega$, with

$$\omega = \text{atan2}(gh - fk, fh + gk), \quad \Omega = \text{atan2}(k, h) \quad (33)$$

Substituting Eq.(32) into Eq.(31) yields

$$\hat{\mathbf{s}}^l = R_z(L - \Omega - \omega) \hat{\mathbf{s}}^P = R_z(L) R_z(-\Omega - \omega) \hat{\mathbf{s}}^P \quad (34)$$

Let

$$\hat{\mathbf{s}}_L = R_z(-\Omega - \omega) \hat{\mathbf{s}}^P = \begin{bmatrix} s_1 & s_2 & s_3 \end{bmatrix}^T \quad (35)$$

Then, using Eq.(34), the minimization problem (30) can be rewritten as

$$\tilde{L}_{\alpha}^*(\mathbf{x}_m) = \arg \min_L [\Gamma_{\alpha}(L)] = \arg \min_L [\varepsilon_{\alpha} \mathbf{A}_{\alpha}(\mathbf{x}_m, L) R_z(L) \hat{\mathbf{s}}_L] \quad (36)$$

By examining the first derivative of $\Gamma_p(L)$, it is found that $\Gamma_p(L)$ admits two stationary points:

$$\begin{aligned} \tilde{L}_{p,1} &= \arctan2\left(-\frac{fs_1 + gs_2 + \tau_1 s_1}{s_2}, \tau_1\right), \\ \tilde{L}_{p,2} &= \arctan2\left(-\frac{fs_1 + gs_2 + \tau_2 s_1}{s_2}, \tau_2\right), \\ \tau_1 &= -\frac{fs_1^2 + gs_1 s_2 - \sigma}{s_1^2 + s_2^2}, \quad \tau_2 = -\frac{fs_1^2 + gs_1 s_2 + \sigma}{s_1^2 + s_2^2}, \\ \sigma &= \sqrt{(1 - f^2)s_1^2 s_2^2 - 2fgs_1 s_2^3 + (1 - g^2)s_2^4} \end{aligned} \quad (37)$$

One of these stationary points corresponds to a local maximum of $\Gamma_p(L)$, while the other corresponds to a local minimum. Which point yields the minimum depends on the values of f , g , and $\hat{\mathbf{s}}_L$. Therefore, the minimizer $\tilde{L}_p$ is selected by direct evaluation:

$$\tilde{L}_p = \arg \min_{L \in \{\tilde{L}_{p,1}, \tilde{L}_{p,2}\}} [\Gamma_p(L)] \quad (38)$$

Similarly, by analyzing the first derivatives of $\Gamma_h(L)$ and $\Gamma_k(L)$, unique minimizers $\tilde{L}_h$ and $\tilde{L}_k$ are obtained in closed form:

$$\tilde{L}_h = \arctan2\left(-g, \tilde{\varepsilon}_h \sqrt{1 - g^2}\right), \quad \tilde{\varepsilon}_h = \text{sign}(\varepsilon_h s_3) \quad (39)$$

$$\tilde{L}_k = \arctan2\left(-\tilde{\varepsilon}_k \sqrt{1 - f^2}, -f\right), \quad \tilde{\varepsilon}_k = \text{sign}(\varepsilon_k s_3) \quad (40)$$

In contrast, for the elements f and g , the functions $\Gamma_f(L)$ and $\Gamma_g(L)$ may exhibit multiple local minima over $L \in [0, 2\pi]$, making it difficult to derive analytical expressions for $\tilde{L}_f$ and $\tilde{L}_g$. Therefore, a mesh search is adopted: $\Gamma_f(L)$ and $\Gamma_g(L)$ are evaluated on a uniform grid of L values in $[0, 2\pi]$, and the grid point that yields the smallest value is selected as $\tilde{L}_f$ or $\tilde{L}_g$, respectively.

Using $\tilde{L}_{\alpha}$ to approximate L_{α}^* and let $\mathbf{x} = [\mathbf{x}_m^T, \tilde{L}_{\alpha}]^T$, problem (28) becomes

$$\begin{aligned} \{\tilde{\alpha}_{\alpha}, \tilde{\beta}_{\alpha}\} &= \arg \min_{\alpha, \beta} [\varepsilon_{\alpha} \mathbf{A}_{\alpha}(\mathbf{x}) \mathbf{a}_{\text{SRP}}^l(\alpha, \beta)] \\ &= \arg \min_{\alpha, \beta} [\varepsilon_{\alpha} \mathbf{A}_{\alpha}(\mathbf{x}) \mathbf{C}_P^l \mathbf{C}_S^P \mathbf{a}_{\text{SRP}}^S(\alpha, \beta)] \\ &= \arg \min_{\alpha, \beta} [\varepsilon_{\alpha} \mathbf{A}_{\alpha}(\mathbf{x}) R_z(\tilde{L}_{\alpha}) R_z(-\Omega - \omega) \mathbf{C}_S^P \mathbf{a}_{\text{SRP}}^S(\alpha, \beta)] \end{aligned} \quad (41)$$

Here, $\mathbf{C}_S^P$ is the DCM that transforms a vector from the sail body-fixed frame $\mathbf{S}$ to the perifocal frame $\mathbf{P}$, and $\mathbf{a}_{\text{SRP}}^S$ is the SRP acceleration expressed in $\mathbf{S}$. Define

$$\Psi_{\alpha} = -\varepsilon_{\alpha} \mathbf{A}_{\alpha}(\mathbf{x}) R_z(\tilde{L}_{\alpha}) R_z(-\Omega - \omega) \mathbf{C}_S^P = \begin{bmatrix} \Psi_1 & \Psi_2 & \Psi_3 \end{bmatrix} \quad (42)$$

where Ψ_α is a 1×3 row vector.

Substituting Eq.(42) into Eq.(41) and assuming a perfectly reflecting sail ($C_1 = 2$, $C_2 = 0$, $C_3 = 0$), closed-form expressions for $\tilde{\alpha}_\alpha$ and $\tilde{\beta}_\alpha$ are obtained as

$$\begin{aligned}\tilde{\alpha}_\alpha &= \arctan\left(\frac{-3\psi + \sqrt{8 + 9\psi^2}}{4}\right), \\ \tilde{\beta}_\alpha &= -\frac{\pi}{2} - \text{atan2}(-\Psi_2, -\Psi_3), \quad \psi = \frac{\Psi_1}{\sqrt{\Psi_2^2 + \Psi_3^2}}\end{aligned}\quad (43)$$

The extreme attainable element rate is then approximated by

$$\dot{\alpha}_{\text{mm}} \approx \dot{\alpha}(x_m, \tilde{L}_\alpha, \tilde{\alpha}_\alpha, \tilde{\beta}_\alpha) \quad (44)$$

Note that the expression in Eq.(43) may yield $\tilde{\beta}_\alpha \notin [-\pi, \pi]$. In implementation, $\tilde{\beta}_\alpha$ is wrapped to the principal interval (e.g., by adding or subtracting 2π as needed) to enforce $\tilde{\beta}_\alpha \in [-\pi, \pi]$.

3.3. Solar-sailing optimal steering law

Once the extreme attainable element rates are available, the proximity quotient Q can be evaluated and the solar-sail steering law is obtained by minimizing its time derivative:

$$\begin{aligned}\mathbf{u}^* &= \arg \min_{\alpha, \beta} \left[\frac{dQ}{dt} \right] = \arg \min_{\alpha, \beta} \left[\frac{\partial Q}{\partial \mathbf{x}} \dot{\mathbf{x}} \right] \\ &= \arg \min_{\alpha, \beta} \left[\frac{\partial Q}{\partial \mathbf{x}} \mathbf{A}(\mathbf{x}) \mathbf{a}_{\text{SRP}}^l(\mathbf{x}, \alpha, \beta) \right] = \arg \min_{\alpha, \beta} \left[\frac{\partial Q}{\partial \mathbf{x}} \mathbf{A}(\mathbf{x}) \mathbf{C}_S^l \mathbf{a}_{\text{SRP}}^s(\alpha, \beta) \right]\end{aligned}\quad (45)$$

where $\mathbf{u}^* = [\alpha^*, \beta^*]^T$. The solution of (45) follows the same structure as that of (41). Define the row vector

$$\tilde{\Psi} = -\frac{\partial Q}{\partial \mathbf{x}} \mathbf{A}(\mathbf{x}) \mathbf{C}_S^l = [\tilde{\Psi}_1, \tilde{\Psi}_2, \tilde{\Psi}_3] \quad (46)$$

The analytical expression for the optimal steering law is:

$$\begin{aligned}\alpha^* &= \arctan\left(\frac{-3\tilde{\psi} + \sqrt{8 + 9\tilde{\psi}^2}}{4}\right), \\ \beta^* &= -\frac{\pi}{2} - \arctan2(-\tilde{\Psi}_2, -\tilde{\Psi}_3), \quad \tilde{\psi} = \frac{\tilde{\Psi}_1}{\sqrt{\tilde{\Psi}_2^2 + \tilde{\Psi}_3^2}}\end{aligned}\quad (47)$$

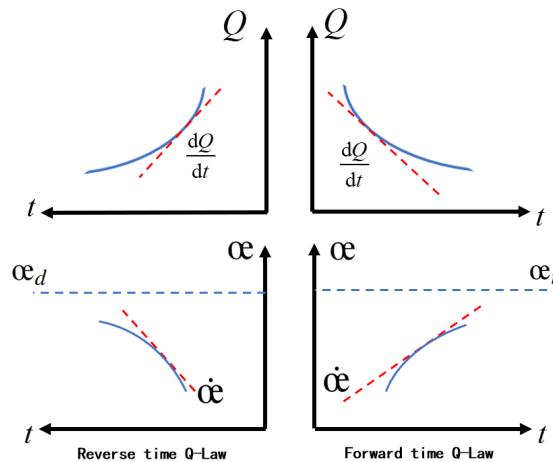


Fig. 5: Illustration of reverse-time Q-Law.

3.4. Reverse-time Q-Law

The forward-time Q-Law introduced above is adapted here into a reverse-time Q-Law. The purpose of the reverse-time formulation is to generate a trajectory that, when propagated from the terminal orbit, converges toward the prescribed departure orbit. Recall that the proximity quotient Q in the forward-time setting is constructed using the terminal (target) orbit, directly propagating the forward-time Q-Law backward in time does not, in general, drive the trajectory back to the departure orbit due to numerical error accumulation during integration. To address this issue, the target-orbit terms in Q are replaced by the corresponding departure-orbit quantities, yielding

$$Q = (1 + P) \sum_{\alpha} W_{\alpha} S_{\alpha} \left(\frac{\alpha - \alpha_d}{\dot{\alpha}_{\text{mm}}} \right)^2 \quad (48)$$

where α_d denotes the elements of the departure orbit.

Compared with the forward-time Q-Law, the reverse-time formulation differs in two aspects. First, the definition of the extreme attainable element rate $\dot{\alpha}_{\text{mm}}$ is reversed: when $\alpha < \alpha_d$, $\dot{\alpha}_{\text{mm}}$ is taken as the minimum value of $\dot{\alpha}$ over α, β , and L , whereas when $\alpha > \alpha_d$, it is taken as the maximum value. Second, since the trajectory is propagated in reverse time, the control is chosen to maximize dQ/dt , as illustrated in Fig. 5. Accordingly, the reverse-time Q-Law is obtained through the substitutions

$$\{L_{\alpha}^*, \alpha_{\alpha}^*, \beta_{\alpha}^*\} = \arg \min_{\alpha, \beta, L} [-\varepsilon_{\alpha} \dot{\alpha}(x, \alpha, \beta)] \quad (49)$$

$$\mathbf{u}^* = \arg \min_{\alpha, \beta} \left[-\frac{dQ}{dt} \right] = \arg \min_{\alpha, \beta} \left[-\frac{\partial Q}{\partial \mathbf{x}} \mathbf{A}(\mathbf{x}) \mathbf{C}_S^T \mathbf{a}_{\text{SRP}}^S(\alpha, \beta) \right] \quad (50)$$

In summary, the reverse-time Q-Law has the same analytical form as the forward-time Q-Law, except that α_t is replaced by α_d , ε_{α} is replaced by $-\varepsilon_{\alpha}$, and the vector $\tilde{\Psi} = -\frac{\partial Q}{\partial \mathbf{x}} \mathbf{A}(\mathbf{x}) \mathbf{C}_S^T$ is replaced by $\tilde{\Psi} = \frac{\partial Q}{\partial \mathbf{x}} \mathbf{A}(\mathbf{x}) \mathbf{C}_S^T$.

3.5. Performance Assessment

To assess the performance of the proposed forward-time and reverse-time Q-Law guidance schemes, three orbit-transfer scenarios are considered. The Keplerian elements of the departure and target orbits, provided in Table 1, are included to illustrate their geometric characteristics. For benchmarking purposes, a recently developed MEE-based solar-sail steering law Yin & Damaren [2025] is also evaluated in Cases B and C. Unless otherwise stated, the element weights are set to unity ($W_{\alpha} = 1$). The minimum allowable periapsis radius is set to $r_p = 6578$ km. The sail area-to-mass ratio is $\sigma = 200$ m²/kg. The Sun and Moon phase angles are initialized to zero at $t = 0$. A transfer is declared successful (i.e., the spacecraft has reached the target orbit, or has returned to the departure orbit in reverse-time propagation) when all of the following conditions are simultaneously satisfied:

$$|p - p_t| < 50 \text{ km}, \quad |f - f_t| < 5 \times 10^{-4}, \quad |g - g_t| < 5 \times 10^{-4}, \quad |h - h_t| < 2 \times 10^{-4}, \quad |k - k_t| < 2 \times 10^{-4}.$$

Fig. 6 shows the forward-time Q-Law results for Case A, which transfers from GTO to a large elliptic orbit with semi-major axis $a = 94,653$ km. This case primarily demonstrates that the proposed guidance law remains effective in the Earth–Moon intermediate region, where the lunar gravitational perturbation is non-negligible. Fig. 7 compares the element histories produced by the proposed forward-time Q-Law against those obtained using the MEE-based steering law of Ref. Yin & Damaren [2025] for Cases B and C. In Case B, the proposed method completes the transfer in 69.1 days, whereas the reference method requires nearly 700 days. In Case C, the corresponding transfer times are 331.7 days and 732.3 days, respectively. These results indicate that the proposed Q-Law yields substantially shorter transfer times, with the largest improvement occurring during the final convergence phase. Finally, the reverse-time Q-Law results for Case C are presented in Fig. 8. The overall duration is slightly longer than that

of the corresponding forward-time transfer. Note that, for the reverse-time simulation, the time of flight is reported over the interval $t \in [0, -350]$ days, with the Sun and Moon phase angles set to zero at $t = 0$.

Table 1: Keplerian orbital elements of departure and target orbits.

Case	Orbit Type	a/km	e	$i/^\circ$	$\Omega/^\circ$	$\omega/^\circ$
A	Departure	24506	0.725	0.1	1	1
	Target	94653	0.868	1	10	213
B	Departure	9223	0.2	0.573	1	1
	Target	26500	0.75	0.01	60	50
C	Departure	24506	0.725	0.05	30	40
	Target	42165	0.001	90	90	50

The convergence robustness of both the forward-time and reverse-time Q-Law is evaluated through a Monte-Carlo simulation. This analysis ensures the control law remains effective across a wide range of orbital geometries and initial conditions. The verification procedure is illustrated in Fig.9. For each randomly generated transfer scenario, the forward-time Q-Law is first tested. If convergence is achieved, the corresponding TOF is used to compute the terminal Sun phase angle, which is then adopted as the initial Sun phase angle for the reverse-time case. If the forward-time case fails to converge, the terminal Sun phase angle is instead estimated assuming a fixed TOF of 300 days for the reverse-time simulation. In each Monte-Carlo loop, both forward-time and reverse-time transfers are evaluated under identical departure and target orbit conditions. The simulation is terminated after 500 trials. The element weights are set to unity: $W_p = W_f = W_g = W_h = W_k = 1$. A transfer is considered converged if the following conditions are simultaneously satisfied within a maximum TOF of 1000 days:

$$|p - p_t| < 300 \text{ km}, \quad |f - f_t| < 1 \times 10^{-3}, \quad |g - g_t| < 1 \times 10^{-3}, \quad |h - h_t| < 1 \times 10^{-3}, \quad |k - k_t| < 1 \times 10^{-3}$$

A total of 500 scenarios are generated and tested. In each trial, the departure and target orbits are generated independently using randomly selected Keplerian elements within prescribed bounds. Specifically, the periapsis radius r_p is first randomly generated within the interval $[7000, 20000]$ km. The apoapsis radius r_a is then randomly assigned under the constraint $r_p < r_a < 9 \times 10^4$ km. The corresponding semi-major axis and eccentricity are computed as $a = (r_a + r_p)/2$, $e = (r_a - r_p)/(r_a + r_p)$. The inclination satisfies $0 < i < 90^\circ$. The right ascension of the ascending node, the argument of periapsis, and the initial Sun phase angle are randomly generated within $[0, 360^\circ]$. The resulting Keplerian elements are subsequently converted into MEEs for compatibility with the Q-Law formulation.

The Monte-Carlo results indicate convergence ratios of 96.25% for the forward-time Q-Law and 99.0% for the reverse-time Q-Law. The few non-convergent cases were observed to converge after minor adjustments to the element weights. This suggests that the uniform weight configuration is adequate for the vast majority of transfer scenarios, although certain combinations of Sun phase angle and orbital geometry exhibit moderate sensitivity to the weight selection.

Overall, the reverse-time Q-Law demonstrates convergence stability comparable to that of the forward-time formulation. No fundamentally invalid transfer scenario was identified when appropriate weights were applied. Although the development of a systematic weight-selection strategy remains an open research topic, it is not the primary focus of this work. Further investigation into optimal weight determination is left for future research.

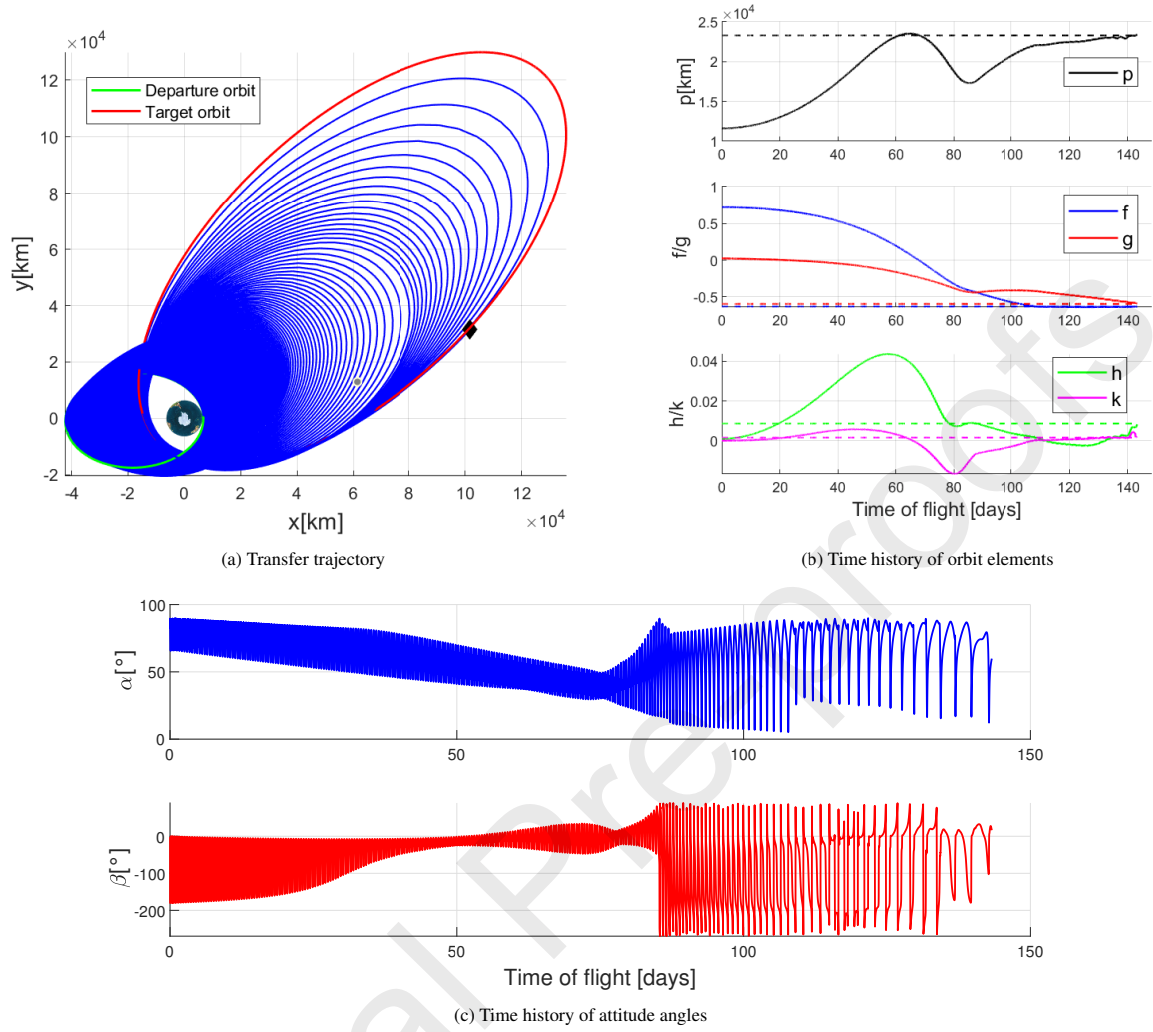
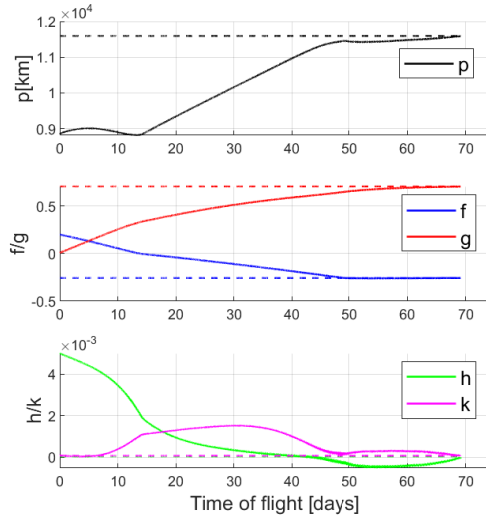
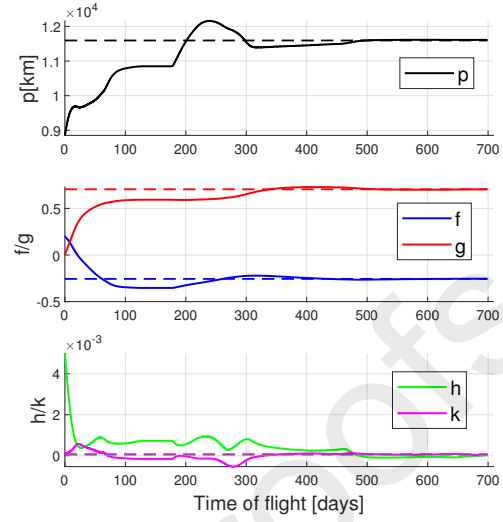


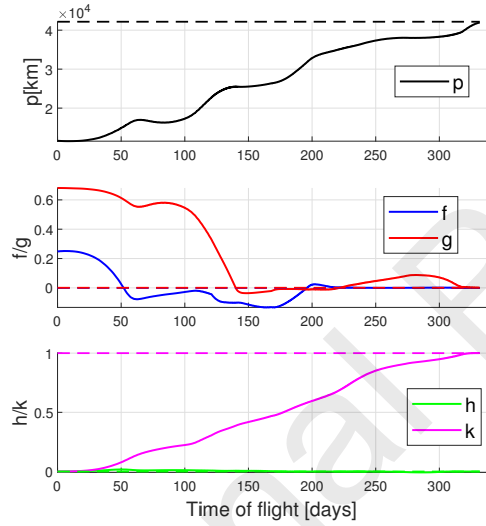
Fig. 6: Results of forward-time Q-Law in case A



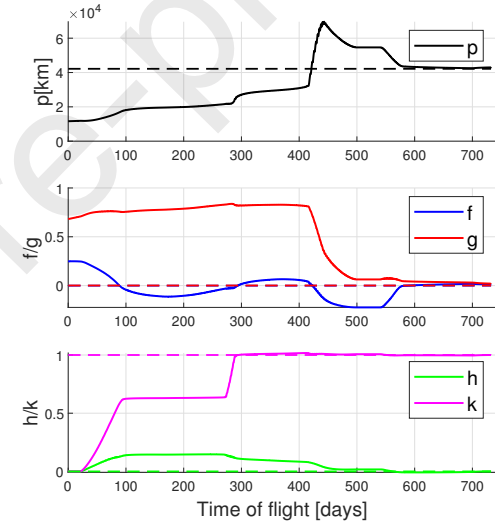
(a) case B: result of this paper



(b) case B: result of Yin&Damaren



(c) case C: result of this paper



(d) case C: result of Yin&Damaren

Fig. 7: Results of forward-time Q-Law in case B and C.

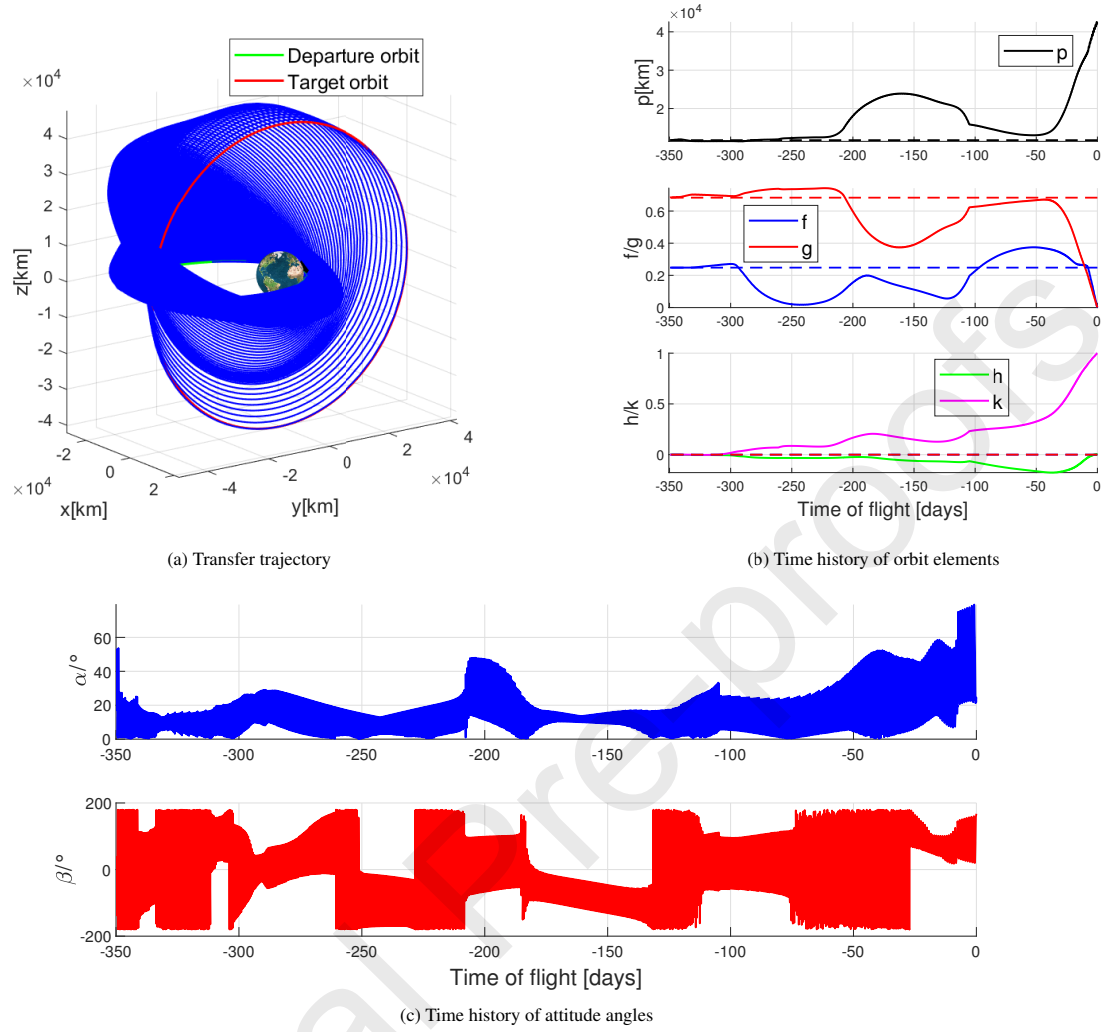


Fig. 8: Results of reverse-time Q-Law in case C

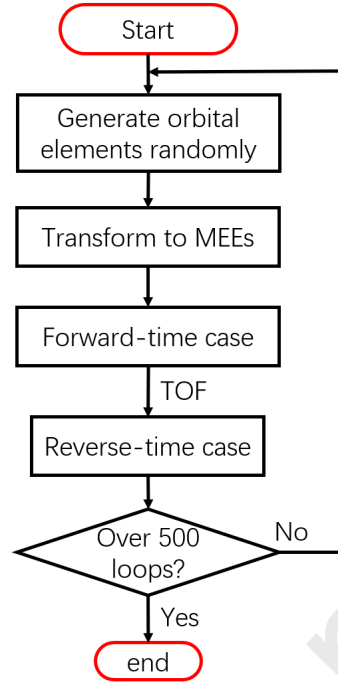


Fig. 9: Flowchart of convergence robustness verification

4. Earth-Moon transfer design

The overall Earth–Moon transfer is decomposed into three phases:

1. **Phase 3 (manifold insertion phase).** In forward time, Phase 3 starts from a preselected state on the stable invariant manifolds associated with the target halo orbit and terminates on the halo orbit itself. A forward–backward multiple-shooting strategy is employed to drive the trajectory from the stable manifold to the target halo orbit.
2. **Phase 2 (energy attenuation phase).** In reverse time, Phase 2 begins at the same preselected manifold state as Phase 3 and ends when the semi-major axis of the resulting orbit is sufficiently small such that the solar-sailing Q-Law performs reliably in the presence of lunar perturbations. A multiple-shooting procedure similar to that in Phase 3 is used to reduce the orbit size (i.e., attenuate the orbital energy).
3. **Phase 1 (departure-orbit regression phase).** In reverse time, Phase 1 starts from the terminal state of Phase 2 and ends at the prescribed departure orbit. The proposed reverse-time Q-Law is applied to steer the spacecraft back to the departure orbit.

The phase-wise trajectories are subsequently patched to form a dynamically consistent end-to-end transfer. The solar-sail-augmented CR3BP dynamics are used in Phases 2 and 3, whereas the MEE-based dynamics are adopted in Phase 1. The assumptions introduced in Secs. 2.2 and 2.4 ensure consistency between these two dynamical models. The Phase 2/Phase 3 patch point is selected *a priori* on the stable invariant manifolds and is shared by both phases. The terminal state of Phase 2 is then transformed to the ECI frame and used as the initial condition for Phase 1. The reverse-time Q-Law used in Phase 1 has been presented above; the detailed procedures for Phases 2 and 3 are described next.

Table 2: Orbital elements of the starting points of Phase 1.

Scenario	a/km	e	$i/^\circ$	$\Omega/^\circ$	$\omega/^\circ$
1	90000	0.85	5	10	10
2	90000	0.85	5	40	70
3	90000	0.85	5	-100	30
4	90000	0.85	5	-20	90
5	90000	0.85	5	10	10
6	90000	0.85	5	-100	20

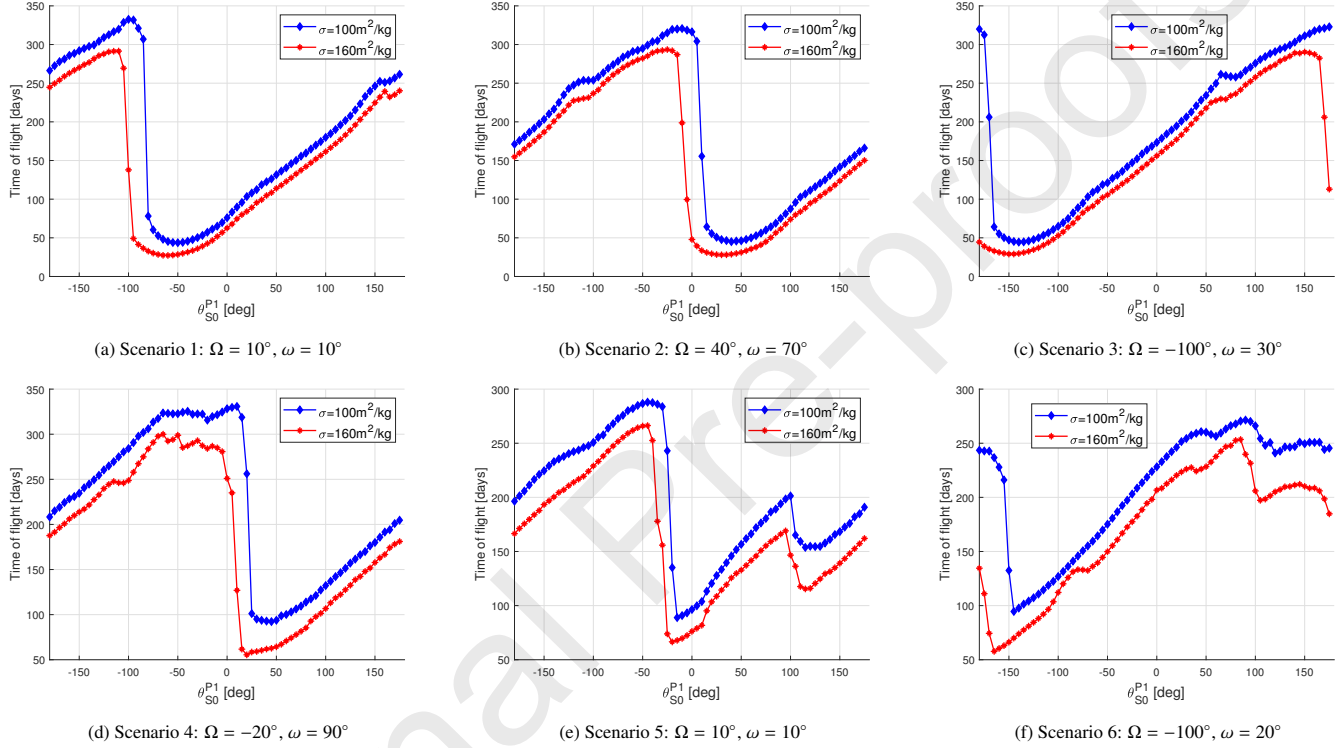


Fig. 10: Phase 1 analysis under various scenarios

4.1. Analysis of Phase 1: Favorable initial condition

The relationship between the time of flight (TOF) and the Phase 1 initial condition is investigated. The Phase 1 initial condition comprises the state vector at the Phase 1 starting point and the initial Sun phase angle in the ECI frame, denoted θ_{S0}^{P1} . Both quantities are obtained by transforming the terminal point of Phase 2 from the barycentric rotating frame to the ECI frame. In the following analysis, the Phase 1 starting point is chosen as a candidate terminal state of Phase 2, and the departure orbit is prescribed as either GEO or GTO. Fig. 10 shows the variation of TOF with respect to θ_{S0}^{P1} . Two area-to-mass ratios are considered, $\sigma = 100 \text{ m}^2/\text{kg}$ and $\sigma = 160 \text{ m}^2/\text{kg}$. Subfigures (a)–(f) correspond to Scenarios 1–6, respectively. As expected, a larger area-to-mass ratio yields a shorter TOF. Across all scenarios, the Sun phase angle associated with the minimum TOF, denoted $\theta_{S0,\min}^*$, consistently occurs ahead of the longitude of periapsis of the start-point orbit, $\tilde{\Omega} = \Omega + \omega$. In addition, the Sun phase angle corresponding to the maximum TOF, $\theta_{S0,\max}^*$, occurs slightly ahead of $\theta_{S0,\min}^*$, and a pronounced jump in TOF is observed between these two angles. These trends indicate that the Sun phase angle inherited from the Phase 2 terminal state is essential and will determine the TOF of Phase 1. Based on these observations, favorable Phase 1 initial Sun phase angles lie in the interval $\theta_{S0}^{P1} \in [\theta_{S0,\min}^*, \theta_{S0,\min}^* + \Delta\theta]$, where $\Delta\theta > 0$ is selected according to the desired TOF requirement for Phase 1.

4.2. Phase2: Energy attenuation phase

The trajectories in Phases 2 and 3 are generated by solving a constrained optimization problem using a forward-backward multiple-shooting strategy. The trajectory is discretized into N nodes, each parameterized by the spacecraft state (position and velocity). The arc between two adjacent nodes is further subdivided into $2N_s$ equal-duration segments, over which the sail attitude angles are defined. The attitude angles are assumed to be piecewise constant within each segment. As illustrated in Fig. 11, the initial time at the Phase 2 starting point is set to zero. All arcs are assigned the same duration, and each arc is divided into segments of equal time span. The attitude angles of the *forward* segments adjacent to the k th node are denoted by $^f\alpha_i^k$ and $^f\beta_i^k$, with $k \in \{2, 3, \dots, N\}$ and $i \in \{1, 2, \dots, N_s\}$. Likewise, the attitude angles of the *backward* segments adjacent to the k th node are denoted by $^b\alpha_i^k$ and $^b\beta_i^k$, with $k \in \{1, 2, \dots, N-1\}$. The design variables are defined as

$$\begin{aligned} \mathbf{P} &= [\mathbf{X}_1^T, \mathbf{X}_2^T, \dots, \mathbf{X}_{N-1}^T, \mathbf{x}_N^T, t_f]^T, \\ \mathbf{X}_k &= [\mathbf{x}_k^T, ^b\alpha_1^k, \dots, ^b\alpha_{N_s}^k, ^b\beta_1^k, \dots, ^b\beta_{N_s}^k, ^f\alpha_1^{k+1}, \dots, ^f\alpha_{N_s}^{k+1}, ^f\beta_1^{k+1}, \dots, ^f\beta_{N_s}^{k+1}]^T \end{aligned} \quad (51)$$

for $k = 1, 2, \dots, N-1$, where $\mathbf{x}_k$ denotes the spacecraft state at the k th node and t_f is the TOF of the phase. Since each arc has duration $t_f/(N-1)$ and is subdivided into $2N_s$ equal segments, the segment time step is

$$\Delta t = \frac{t_f}{2N_s(N-1)} \quad (52)$$

Within each arc, the state propagated backward from node k and the state propagated forward from node $k+1$ are required to match at the arc midpoint:

$$\mathbf{F}_b\left(\mathbf{x}_k, \frac{t_f}{2(N-1)}\right) - \mathbf{F}_f\left(\mathbf{x}_{k+1}, \frac{t_f}{2(N-1)}\right) = \mathbf{0} \quad (53)$$

Here, $\mathbf{F}_b(\mathbf{x}_k, \frac{t_f}{2(N-1)})$ denotes the state obtained by integrating backward from $\mathbf{x}_k$ over a time interval $\frac{t_f}{2(N-1)}$ using the piecewise-constant attitude angles $\{^b\alpha_i^k, ^b\beta_i^k\}_{i=1}^{N_s}$, whereas $\mathbf{F}_f(\mathbf{x}_{k+1}, \frac{t_f}{2(N-1)})$ denotes the state obtained by integrating forward from $\mathbf{x}_{k+1}$ over the same interval using $\{^f\alpha_i^{k+1}, ^f\beta_i^{k+1}\}_{i=1}^{N_s}$. In addition, the initial-node constraint is imposed as

$$\mathbf{x}_1 - \mathbf{x}_0 = \mathbf{0} \quad (54)$$

where $\mathbf{x}_0$ is the manifold point selected *a priori* from the stable invariant manifold.

Given the critical role of the Phase 2 terminal state, the terminal Sun phase angle of Phase 2 is constrained to lie close to the favorable initial Sun phase angle for Phase 1. Accordingly, the terminal constraints are imposed as

$$a_f < 9.5 \times 10^4 \text{ km}, \quad (55)$$

$$r_{p,f} > 7000 \text{ km}, \quad (56)$$

$$\theta_{S0}^{P1} - \tilde{\Omega}_f < \Delta\theta' \quad (57)$$

Here, a_f , $r_{p,f}$, θ_{S0}^{P1} , and $\tilde{\Omega}_f$ denote the semi-major axis, periaapsis radius, Sun phase angle, and longitude of periaapsis, respectively, of the Phase 1 initial state in the ECI frame. This Phase 1 initial state is obtained by transforming the Phase 2 terminal state from the barycentric rotating frame to the ECI frame. The parameter $\Delta\theta'$ prescribes the maximum allowable deviation of θ_{S0}^{P1} from $\tilde{\Omega}_f$.

The periaapsis-radius lower bound is set to 7000 km to provide margin such that, at the beginning of Phase 1, the periaapsis radius will not drop below the minimum allowable value of 6578 km under subsequent Q-Law guidance. Finally, the objective function is set to a constant, i.e., $J = 0$, and the solution is sought by satisfying the above constraints.

To summarize, Phase 2 seeks a feasible solution that satisfies the matching and boundary constraints in (53)–(57). An initial guess is constructed by discretizing a trajectory generated with the locally optimal steering law proposed in Macdonald & McInnes [2005a,b]. The key idea of this law is to maximize the projection of the SRP acceleration onto the velocity direction, i.e.,

$$\mathbf{u}^* = \arg \min_{\alpha, \beta} \left[-\mathbf{v}^R \mathbf{C}_S^R \mathbf{a}_{\text{SRP}}^S(\alpha, \beta) \right] \quad (58)$$

where $\mathbf{C}_S^R$ is the direction-cosine matrix that transforms a vector from the solar-sail body-fixed frame to the barycentric rotating frame, $\mathbf{v}^R$ represents the velocity vector expressed in the barycentric rotating frame.

In practice, the analytical steering law is obtained from Eq.(47) by replacing $\tilde{\Psi}$ with $\Psi_v = \mathbf{v}^R \mathbf{C}_S^R$. The initial-guess trajectory is generated by propagating backward from the Phase 2 initial state under this locally optimal law until the semi-major axis falls below 9.5×10^4 km. For this initial guess, however, the terminal Sun phase angle is typically far from the favorable Phase 1 range. Moreover, we observe that $\theta_{\text{S0}}^{\text{P1}}$ tends to be hysteretic to $\tilde{\Omega}_f$, whereas $\theta_{\text{S0},\text{min}}^*$ is consistently ahead of $\tilde{\Omega}_f$. Consequently, the constraint in (57) is introduced to bias $\theta_{\text{S0}}^{\text{P1}}$ toward $\theta_{\text{S0},\text{min}}^*$, thereby reducing the Phase 1 time of flight. Because it is sometimes the case that the solver converges to an infeasible point, a continuation strategy with respect to $\Delta\theta'$ is adopted to improve robustness. Specifically, $\Delta\theta'$ is initialized to the value implied by the initial-guess trajectory and is then gradually decreased over successive solves. The smallest attainable $\Delta\theta'$ corresponds to the best Phase 2 solution.

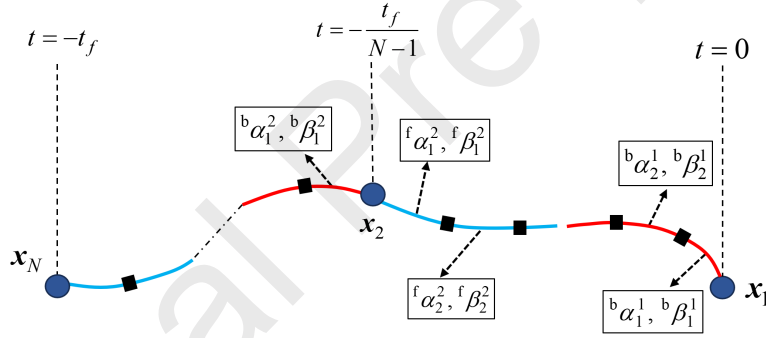


Fig. 11: Diagram of nodes, forward segments (blue color) and backward segments (red color) in Phase 2.

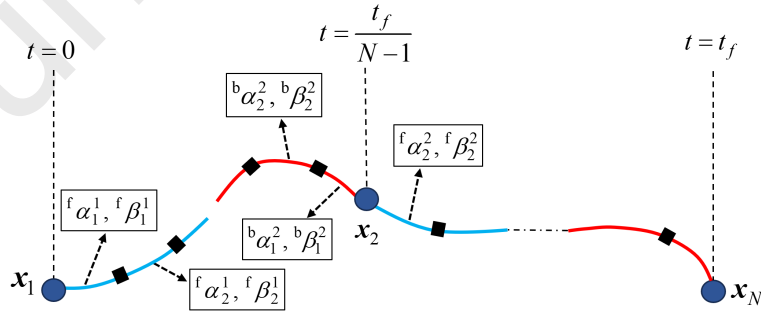


Fig. 12: Diagram of nodes, forward segments (blue color) and backward segments (red color) in Phase 3.

4.3. Phase 3: Manifold insertion phase

Recall that the Phase 2 initial state is selected *a priori* on the stable invariant manifold associated with the target halo orbit, and Phases 2 and 3 are patched at this manifold point. Consequently, Phase 2 and Phase 3 share an identical starting state, and the corresponding initial Sun phase angles are also identical, i.e., $\theta_{\text{S0}}^{\text{P3}} = \theta_{\text{S0}}^{\text{P2}}$.

As shown in Fig. 12, the multiple-shooting formulation for Phase 3 is a minor modification of that used in Phase 2. In Phase 3, the shooting time domain is $t \in [0, t_f]$, whereas in Phase 2 it is $t \in [0, -t_f]$. Note that the actual TOFs of Phases 2 and 3 are generally different; here, t_f denotes the time horizon introduced in the multiple-shooting transcription.

The decision variables for Phase 3 are defined as

$$\begin{aligned} \mathbf{P} &= [\mathbf{X}_1^T, \mathbf{X}_2^T, \dots, \mathbf{X}_{N-1}^T, \mathbf{x}_N^T, t_f]^T, \\ \mathbf{X}_k &= [\mathbf{x}_k^T, {}^f\alpha_1^k, \dots, {}^f\alpha_{N_s}^k, {}^f\beta_1^k, \dots, {}^f\beta_{N_s}^k, {}^b\alpha_1^{k+1}, \dots, {}^b\alpha_{N_s}^{k+1}, {}^b\beta_1^{k+1}, \dots, {}^b\beta_{N_s}^{k+1}]^T \end{aligned} \quad (59)$$

for $k = 1, 2, \dots, N-1$, where $\mathbf{x}_k$ is the spacecraft state at the k th node. The segment duration remains $\Delta t = t_f/(2N_s(N-1))$, and the attitude angles are assumed piecewise constant over each segment. The midpoint matching constraint for Phase 3 enforces continuity between the forward propagation from node k and the backward propagation from node $k+1$:

$$\mathbf{F}_f\left(\mathbf{x}_k, \frac{t_f}{2(N-1)}\right) - \mathbf{F}_b\left(\mathbf{x}_{k+1}, \frac{t_f}{2(N-1)}\right) = \mathbf{0} \quad (60)$$

Finally, the boundary conditions are imposed as

$$\mathbf{x}_1 - \mathbf{x}_0 = \mathbf{0}, \quad (61)$$

$$\mathbf{x}_N - \mathbf{x}_{\text{Halo}} = \mathbf{0} \quad (62)$$

where $\mathbf{x}_{\text{Halo}}$ is the prescribed terminal state on the target halo orbit. Therefore, Phase 3 is formulated as a feasibility problem: find $\mathbf{P}$ such that (60)–(62) are satisfied. The initial guess is constructed by discretizing the manifold trajectory segment connecting the selected manifold point $\mathbf{x}_0$ to the halo orbit.

As shown in Fig. 13, the overall Earth–Moon transfer design procedure can be summarized as follows:

1. **Parameter initialization.** Initialize the problem parameters, including the solar-sail area-to-mass ratio σ , the initial Sun phase angle for Phase 2 (θ_{S0}^{P2}), the terminal and initial states on the halo orbit and the associated stable manifold ($\mathbf{x}_0, \mathbf{x}_{\text{Halo}}$), the numbers of nodes and segments in Phases 2 and 3 (N and $2N_s$), the Q-Law weights W_{α} , and the orbital elements of the departure orbit (GEO or GTO).
2. **Phase 3 solution.** Construct an initial guess for Phase 3 by discretizing the manifold trajectory and solve the Phase 3 multiple-shooting problem. If the solver fails to converge to a feasible solution, the corresponding Earth–Moon transfer is deemed infeasible.
3. **Phase 2 solution with continuation.** Construct an initial guess for Phase 2 using the locally optimal steering law and solve the Phase 2 problem with a continuation strategy on $\Delta\theta'$. The parameter $\Delta\theta'$ is gradually reduced until no convergent solution can be obtained. The last converged solution in this continuation sequence is taken as the optimal Phase 2 result.
4. **Phase 1 design via reverse-time Q-Law.** Transform the Phase 2 terminal state from the barycentric rotating frame to the ECI frame. The corresponding Moon and Sun phase angles are also converted using the Phase 2 terminal time. This transformed state is used as the initial condition for Phase 1, which is then propagated backward in time under the reverse-time Q-Law until the spacecraft reaches the departure orbit. The convergence criteria employed in Sec. 3.5 are applied here.
5. **Synthesis of the three phases.** Finally, the trajectories obtained in Phases 1–3 are concatenated to yield the complete Earth–Moon transfer.

The use of reverse time in Phases 1 and 2 is a deliberate design choice to avoid a two-point boundary value problem in Phase 2. Recall that the Q-Law regulates the terminal orbital elements except for the true longitude; thus, the terminal state of Phase 1 is not fully determined until Phase 1 has been propagated. If a forward-time Q-Law were used in Phase 1, the Phase 2 initial state would coincide with the Phase 1 terminal state. In that case, Phases 2 and 3 together would constitute a fixed-endpoint transfer, for which constructing a suitable initial guess and achieving convergence is significantly more challenging. By contrast, when reverse time is employed in both Phases 1 and 2, the Phase 2 terminal state is not prescribed a priori, and Phase 2 becomes a more tractable feasibility problem with flexible endpoint conditions, for which a convergent solution can reliably be obtained. Furthermore, the Phase 3 trajectory is nearly ballistic, as the spacecraft follows the invariant manifold toward the halo orbit. As a result, the proposed three-phase framework promotes robust convergence in both Phase 2 and Phase 3 while enabling an effective synthesis of the overall Earth–Moon transfer.

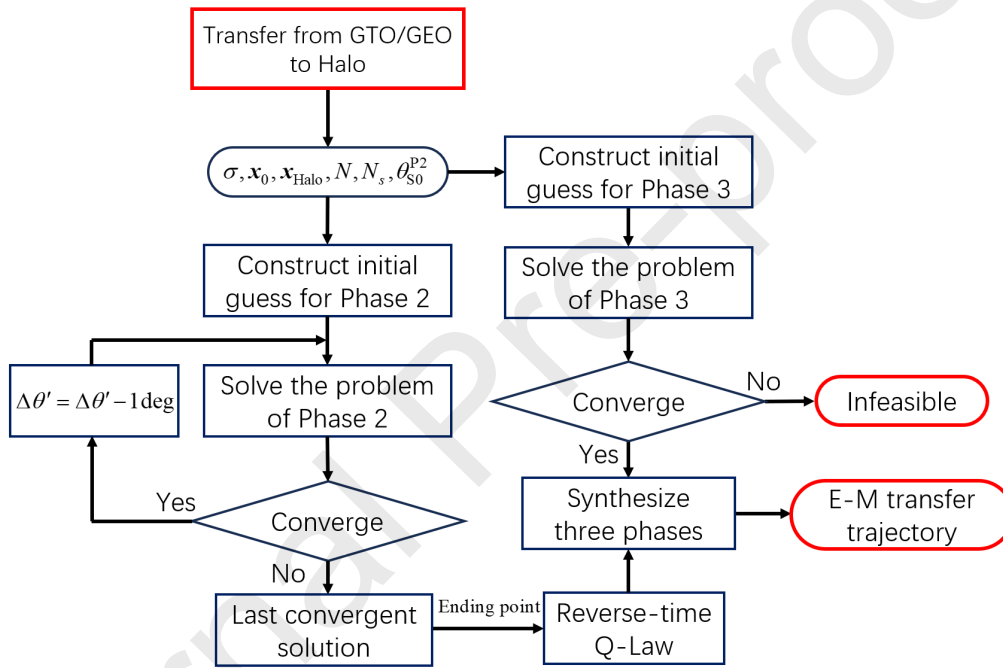


Fig. 13: Flowchart of E-M transfer design.

5. Results and discussion

The design results are presented and analyzed in this section. The Phase 3 and Phase 2 optimization problems are solved using the `fmincon` routine in MATLAB with the SQP algorithm and parallel computing is employed to reduce computational cost. The main optimization options are specified as: `MaxIterations` = 100, `StepTolerance` = 10^{-8} , `ConstraintTolerance` = 10^{-8} . In the multiple-shooting scheme, the numbers of nodes and segments are chosen as $N = 5$ and $N_s = 3$ for Phase 3, and $N = 16$ and $N_s = 3$ for Phase 2. The $L1/L2$ halo orbits are selected as the terminal orbits of the Earth–Moon transfers. The $L1$ halo orbit is a northern-family solution with out-of-plane amplitude $A_z = 2.97 \times 10^4$ km, whereas the $L2$ halo orbit is a southern-family solution with $A_z = 8000$ km. The solar-sail area-to-mass ratio is varied from $\sigma = 100$ m²/kg to $\sigma = 200$ m²/kg. The corresponding maximum SRP accelerations are approximately 0.9–1.8 mm/s², which are of the same order as those adopted in [Petropoulos, 2004; Macdonald & McInnes, 2005a]. For simplicity, the initial Moon phase angle in the ECI frame is set to zero at $t = 0$ (i.e., $\theta_{M0} = 0$), so that the initial Sun phase angle in the ECI frame, θ_{s0}^{p2} , is equal to its value in the rotating frame. The preselected starting

points $\mathbf{x}_0$ on the stable invariant manifolds of the halo orbits for Phases 2 and 3 are listed in Table 3, while the terminal points $\mathbf{x}_{\text{Halo}}$ of Phase 3 on the halo orbits are listed in Table 4. The results of GEO-Halo and GTO-Halo are exhibited and discussed first, then the effect of Earth shadowing on the solar sail in Phase 1 are discussed.

Table 3: Starting point of Phase 2 and Phase 3.

Target Halo orbit	State vector in rotating frame
L1	[0.0916740809 -0.5164936805 0.05997973223 0.9410436426 0.1007625384 -0.01475586973]
L2	[0.4594374144 -0.3146672 0.0086632858291 0.66496087590 0.49090013623 0.050303954256]

Table 4: Terminal points of Phase 3.

Target Halo orbit	State vector in rotating frame
L1	[0.82551969555 0 -0.077370263460 0 0.19072264695 0]
L2	[1.1736717745 -0.045115436833 -0.026203114660 -0.041722914745 -0.13922795073 0.018567084654]

5.1. From GEO to Halo orbit

In this case, GEO is selected as the departure orbit. The Keplerian elements of GEO are $a = 42,165$ km, $e = 0$, and $i = 0$. The Q-Law weights in Phase 1 are chosen as $W_p = 11$, $W_f = 2$, $W_g = 1$, $W_h = 1$, and $W_k = 1$. Fig. 14 presents a GEO–L1 halo transfer for a solar sail with area-to-mass ratio $\sigma = 100$ m²/kg and initial Sun phase angle $\theta_{S0}^{P2} = 200^\circ$. The total time of flight (TOF) is 165.43 days, most of which is spent in Phase 1. Subfigures (a) and (b) show the trajectory in the Earth–Moon barycentric rotating frame and the ECI frame, respectively, with different colors indicating the three phases. The red and magenta dots indicate the patch points between adjacent phases. Subfigure (c) displays the time histories of the attitude angles in Phases 3, 2, and 1 from top to bottom; the horizontal axis is continuous across all three phases, reflecting the continuity of the TOF. The Phase 3 trajectory closely follows the invariant manifold, as indicated by the attitude history: the cone angle remains close to 90° , so the effective SRP acceleration is nearly zero in this phase. Subfigure (d) shows the evolution of the orbital elements during Phase 1. Similarly, Fig. 15 shows a GEO–L2 halo transfer for a solar sail with $\sigma = 200$ m²/kg and initial Sun phase angle $\theta_{S0} = 180^\circ$. In this case, the total TOF is 123.92 days. The main difference from the GEO–L1 case is that, in Phase 3, the spacecraft performs a close flyby of the Moon, which causes the trajectory geometry in the inertial frame to appear compressed.

Table 5: GEO-Halo transfer results.

σ [m ² /kg]	Target Halo orbit	$\theta_{S0}^{P2}/^\circ$	Terminal $\Delta\theta'/^\circ$	Phase 1[days]	Phase 2[days]	Phase 3[days]	Total ToF[days]
100	L1	200	39	107.18	39.15	19.10	165.43
	L2	180	33	98.45	37.64	19.03	155.13
120	L1	200	36	99.39	38.24	19.10	156.74
	L2	180	30	91.27	37.12	19.03	147.42
140	L1	200	33	92.73	37.69	19.10	149.51
	L2	180	26	84.33	37.15	18.98	140.45
160	L1	200	29	85.87	37.47	19.10	142.44
	L2	180	24	80.46	36.29	19.03	135.77
180	L1	200	26	81.56	37.07	19.10	137.73
	L2	180	21	75.27	35.98	19.03	130.28
200	L1	200	23	77.75	36.60	19.10	133.45
	L2	180	18	68.83	36.06	19.03	123.92

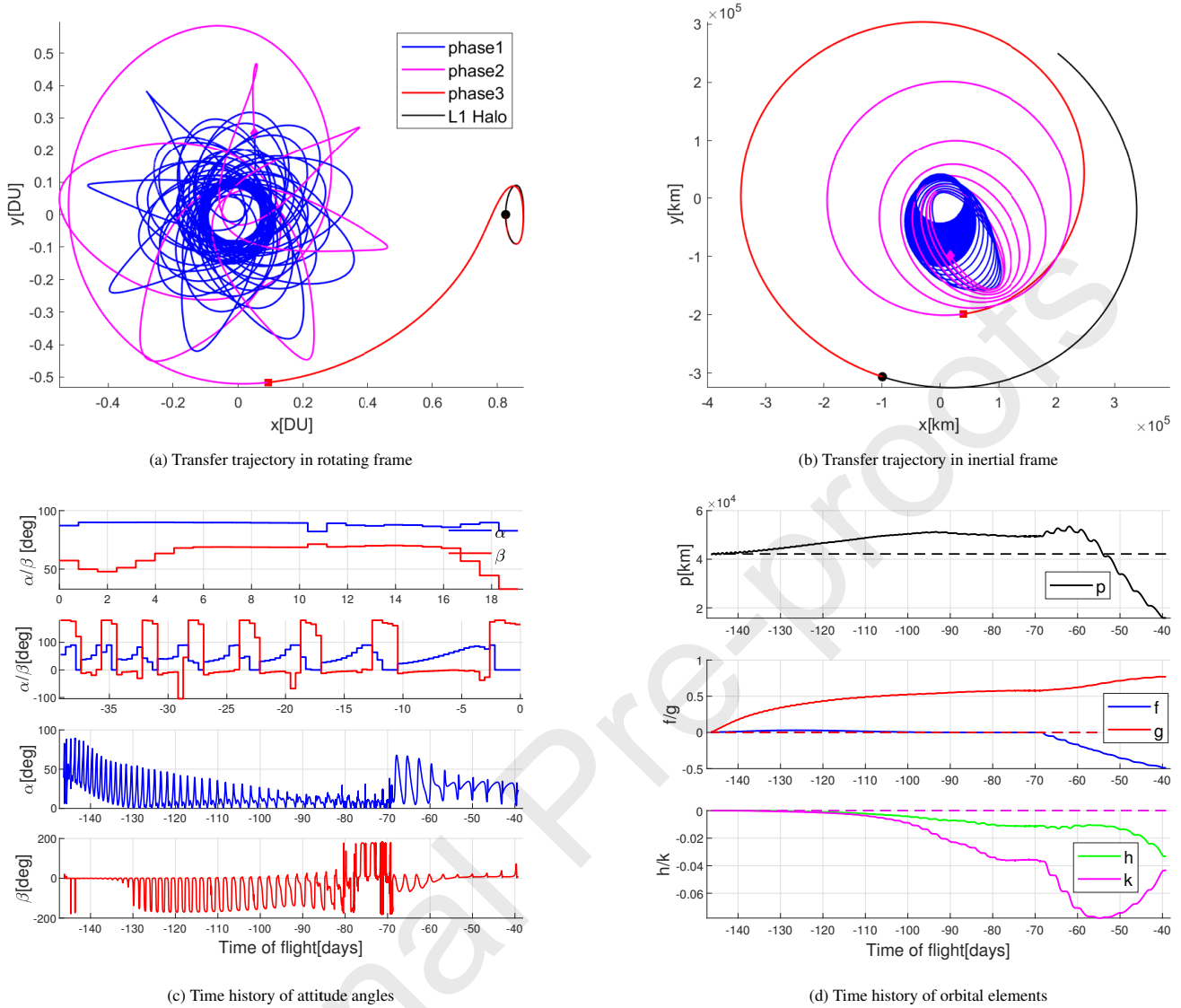


Fig. 14: GEO-L1Halo transfer result ($\sigma = 100 \text{ m}^2/\text{kg}$, $\theta_{S0}^{P2} = 200^\circ$)

To assess the influence of the area-to-mass ratio on the total time of flight (TOF), GEO-L1 halo and GEO-L2 halo transfers are computed for $\sigma \in \{100, 120, 140, 160, 180, 200\} \text{ m}^2/\text{kg}$. As summarized in Tab 5, the initial Sun phase angle is fixed at 200° for all GEO-L1 halo transfers and at 180° for all GEO-L2 halo transfers. The results exhibit a clear trend: the total TOF decreases as the area-to-mass ratio increases. The TOF of Phase 1 decreases markedly with increasing area-to-mass ratio. This behavior is closely related to the terminal value of $\Delta\theta'$, the continuation result associated with the best Phase 2 solution: as σ increases, the terminal $\Delta\theta'$ decreases, which causes the Phase 1 TOF decrease as shown in Fig.10. In contrast, the TOFs of Phase 3 show little variation across the different cases because the Phase 3 endpoints are fixed and the trajectory in this phase is nearly ballistic. For the GEO-L1 halo cases, $\Delta\theta'$ is initialized at 76° and gradually reduced during continuation to the terminal value listed in Tab 5. For the GEO-L2 halo cases, the initial value is 72° . In all scenarios in Tab 5, the initial Sun phase angle of Phase 1, θ_{S0}^{P1} , obtained from the Phase 2 initial guess, initially lags the corresponding longitude of periapsis $\tilde{\Omega}$. The continuation process with respect to $\Delta\theta'$ reduces this difference, but the terminal $\Delta\theta'$ values indicate that θ_{S0}^{P1} remains hysteretic with respect to $\tilde{\Omega}$.

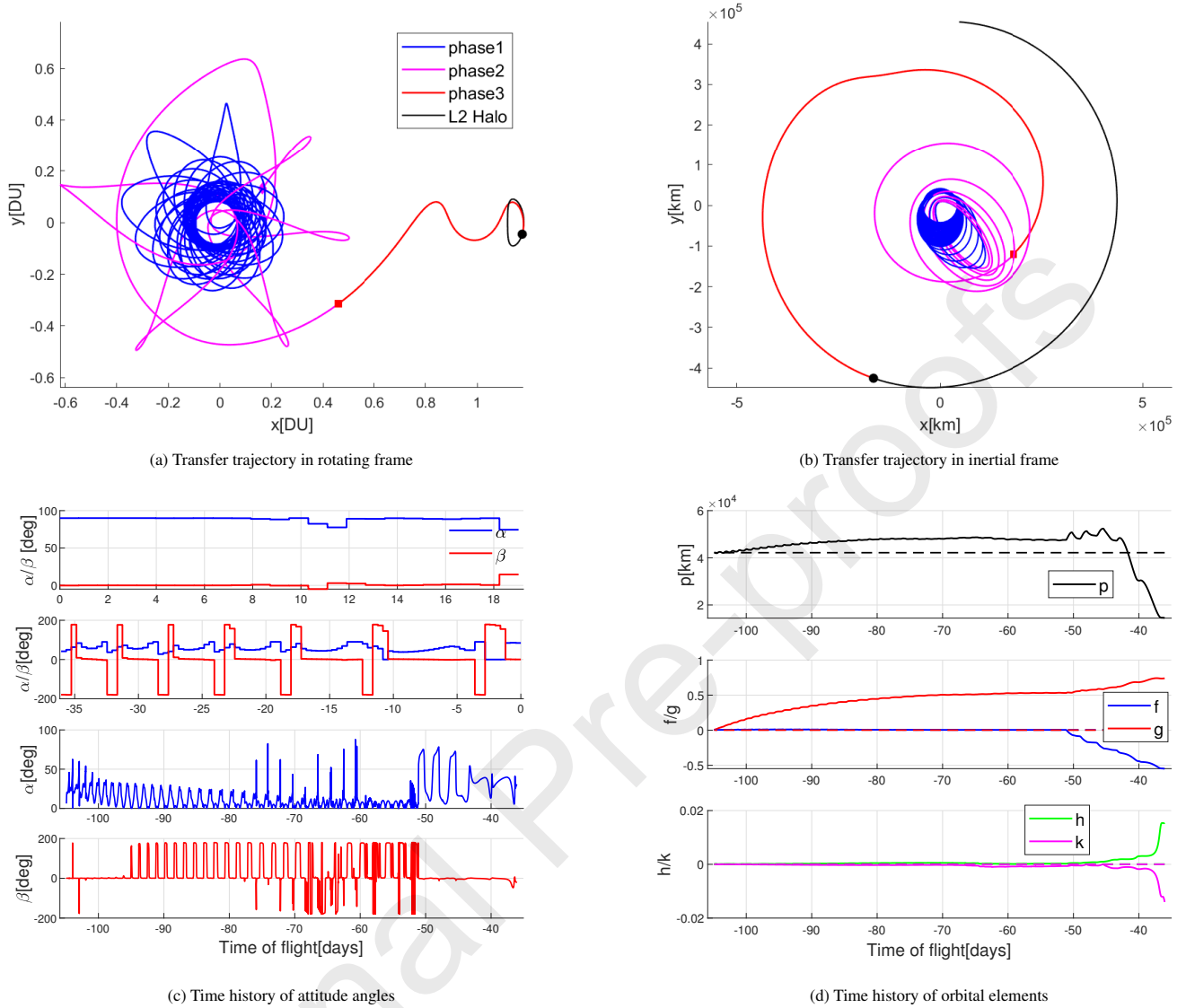


Fig. 15: GEO-L2Halo transfer result ($\sigma = 200 \text{ m}^2/\text{kg}$, $\theta_{S0}^{P2} = 180^\circ$)

5.2. From GTO to Halo orbit

The Earth–Moon transfers using GTO as the departure orbit are also investigated. The Keplerian elements of the GTO are $a = 24,509 \text{ km}$, $e = 0.725$, $i = 0.01^\circ$, $\Omega = 50^\circ$, and $\omega = 50^\circ$. In Phase 1, the Q-Law weights are set to $W_p = 11$, $W_f = 1$, $W_g = 4$, $W_h = 1$, and $W_k = 1$. Fig. 16 shows a representative GTO–L1 halo transfer with $\theta_{S0}^{P2} = 230^\circ$ and $\sigma = 160 \text{ m}^2/\text{kg}$. The total TOF is 130.25 days, which is shorter than that of the GEO–L1 halo transfer with the same area-to-mass ratio. The primary differences arise in the terminal value of $\Delta\theta'$ and in the Phase 1 TOF. A summary of GTO–halo transfer results for various area-to-mass ratios is provided in Tab 6. For all GTO–L1 halo transfers, the initial Sun phase angle for Phase 2 is fixed at $\theta_{S0}^{P2} = 230^\circ$, whereas for the GTO–L2 halo transfers it is fixed at $\theta_{S0}^{P2} = 50^\circ$. Similar to the GEO–halo cases, the TOFs of Phases 2 and 3 are relatively insensitive to the area-to-mass ratio, while the Phase 1 TOF decreases significantly as σ increases. Furthermore, the terminal value of $\Delta\theta'$ is influenced by both the area-to-mass ratio and the initial Sun phase angle of Phase 2, θ_{S0}^{P2} . Among the tested values, a favorable choice is $\theta_{S0}^{P2} \approx 230^\circ$, which yields the smallest terminal $\Delta\theta'$ and thus the most efficient Phase 1 transfer.

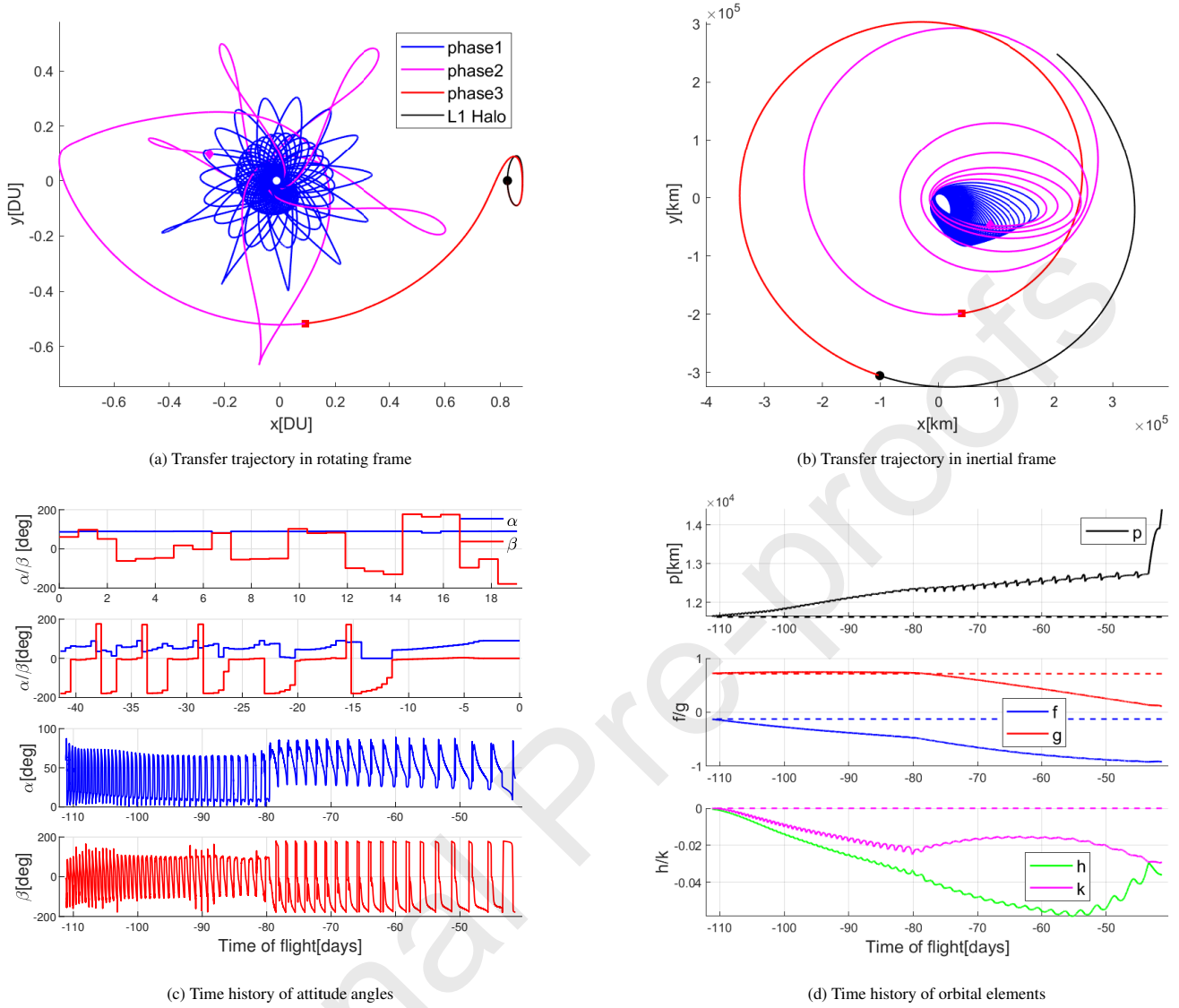
Fig. 16: GTO-L1Halo transfer result ($\sigma = 160 \text{ m}^2/\text{kg}$, $\theta_{S0}^{P2} = 230^\circ$)

Table 6: GTO-Halo transfer results.

$\sigma[\text{m}^2/\text{kg}]$	L1/L2 halo	$\theta_{S0}^{P2}[\text{deg}]$	Terminal $\Delta\theta'[\text{deg}]$	Phase 1[days]	Phase 2[days]	Phase 3[days]	Total ToF[days]
100	L1	230	28	103.57	43.40	19.07	166.04
	L2	50	53	162.47	38.48	19.56	220.52
120	L1	230	24	94.77	42.07	19.07	155.91
	L2	50	44	152.70	38.44	19.67	210.82
140	L1	230	19	78.31	41.69	19.07	147.42
	L2	50	40	134.78	37.29	19.73	191.81
160	L1	230	16	69.78	41.39	19.07	130.25
	L2	50	37	126.96	36.58	19.78	183.31
180	L1	230	12	62.61	41.38	19.07	123.06
	L2	50	33	118.47	36.58	19.81	174.87
200	L1	230	9	57.12	41.24	19.07	117.43
	L2	50	31	114.82	36.17	19.83	170.82

5.3. Effect of Eclipse

If the sail enters an eclipse region (i.e., within the shadow of the Earth or the Moon), solar radiation pressure vanishes and the spacecraft motion becomes purely ballistic. During an Earth–Moon transfer, eclipses occur frequently in near-Earth space. As the spacecraft moves away from the Earth, the eclipse frequency decreases and its influence becomes negligible in the intermediate region. When approaching the Moon, the eclipse frequency gets higher again. In the GEO/GTO–to–L1/L2 halo transfer considered in this study, the spacecraft remains far from both the Earth and the Moon during Phase 2, so eclipse effects are insignificant in that phase. In Phase 1, however, the sail frequently passes through the Earth’s shadow. In Phase 3, lunar eclipses may occur, but the required maneuvering effort is relatively small because the trajectory follows invariant-manifold structures and requires only minor corrections. Therefore, the present analysis focuses on the impact of Earth eclipses during Phase 1.

Under the assumption of parallel sunlight adopted throughout this paper, a simplified cylindrical shadow model is employed for the Earth in near-Earth space. Within the shadow region, the solar sail produces no thrust. Two representative transfer scenarios are considered. The Keplerian elements of the departure and target orbits are listed in Table 7, where the departure orbit is GTO and the target orbit is selected near the terminal state of Phase 2. Both forward-time and reverse-time transfers are examined. The element weights are given in Table 8, and multiple sail area-to-mass ratios are evaluated. Fig. 17 presents representative forward-time results for an area-to-mass ratio of $\sigma=200 \text{ m}^2/\text{kg}$ and an initial Sun phase angle $\theta_{S0}^{P1}=65^\circ$. Subfigures (a) and (c) include eclipse effects, whereas (b) and (d) neglect them. When eclipse is modeled, the TOF increases by approximately 5 days. However, the overall trajectory geometry and the histories of the orbital elements remain nearly unchanged, indicating that the eclipse primarily introduces a modest delay rather than qualitative trajectory modification. Table 9 summarizes results under various combinations of area-to-mass ratio and initial Sun phase angle. In most cases, incorporating eclipse slightly increases the TOF, although the difference is small relative to the total transfer duration; in several cases, the deviation is less than one day. Interestingly, for the specific case $\sigma=150 \text{ m}^2/\text{kg}$ and $\theta_{S0}^{P1}=-105^\circ$, the TOF is marginally shorter when eclipse is included, confirming that the Q-Law guidance is not strictly time-optimal. Overall, eclipse effects tend to lengthen the transfer time while the trajectories and the orbital elements changing histories show few differences with the cases without eclipse. Finally, Table 10 lists the Phase 1 TOFs for GTO–halo transfers with eclipse, for comparison with Table 6 (without eclipse). The TOF deviations in Phase 1 range from 1.29 to 11.97 days. Note that, since Phase 1 is designed under reverse-time propagation and eclipse effects in Phases 2 and 3 are neglected, the updated Earth–Moon transfer solution can be obtained by replacing the original Phase 1 result (without eclipse) with the corresponding result including eclipse, while keeping Phases 2 and 3 unchanged.

Table 7: Orbital elements of departure and target orbits

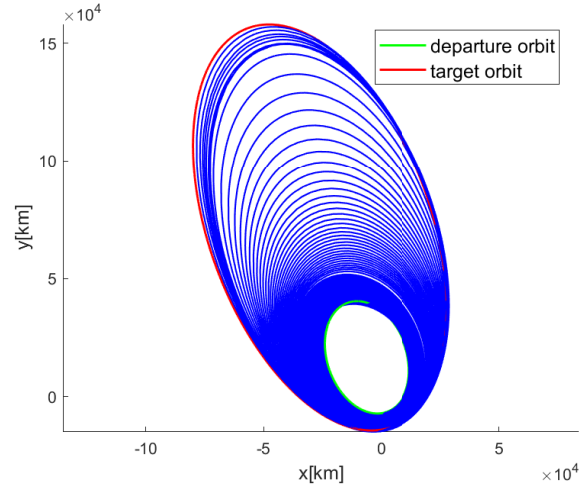
scenario	orbit type	a [km]	e	i [deg]	Ω [deg]	ω [deg]
1	departure	24506	0.725	0.01	0.1	-70
	target	9e4	0.85	5	-100	30
2	departure	24506	0.725	0.01	50	60
	target	9e4	0.85	5	100	30

Table 8: Weights for orbital elements in various cases

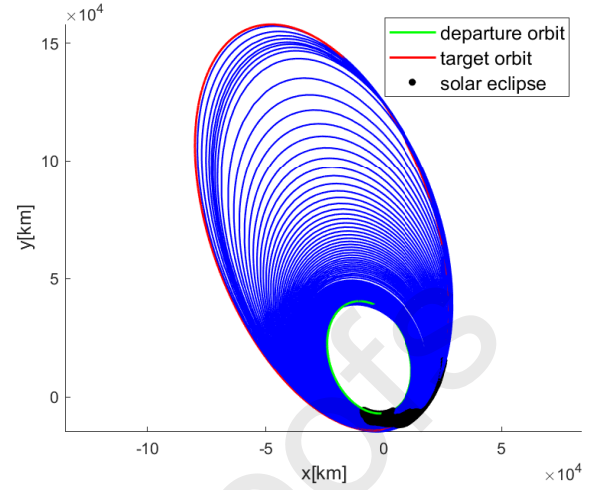
scenario	Time flow	W_a	W_e	W_i	W_Ω	W_ω
1	reverse	1	1	8	10	4
	forward	5	1	1	1	1
2	reverse	14	1	8	1	1
	forward	2	1	1	1	1

Table 9: Summary of the results under combined conditions

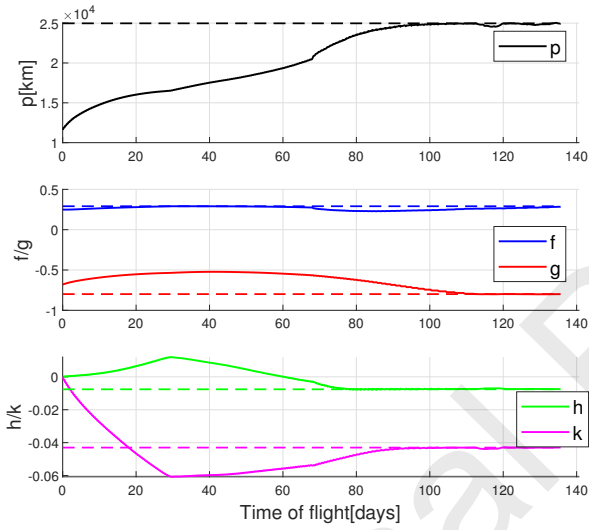
scenario	with eclipse	time flow	σ [m ² /kg]	θ_{S0} [deg]	TOF [days]
1	yes	reverse	200	-105	112.2
	no	reverse	200	-105	111.7
	yes	reverse	150	-105	116.6
	no	reverse	150	-105	117.7
	yes	forward	200	65	140.7
	no	forward	200	65	135.6
	yes	forward	100	65	174.6
	no	forward	100	65	173.1
2	yes	reverse	200	-105	152.4
	no	reverse	200	-105	150.2
	yes	reverse	150	-105	168.7
	no	reverse	150	-105	164.1
	yes	forward	200	-55	130.6
	no	forward	200	-55	130.0
	yes	forward	100	-55	199.3
	no	forward	100	-55	196.8



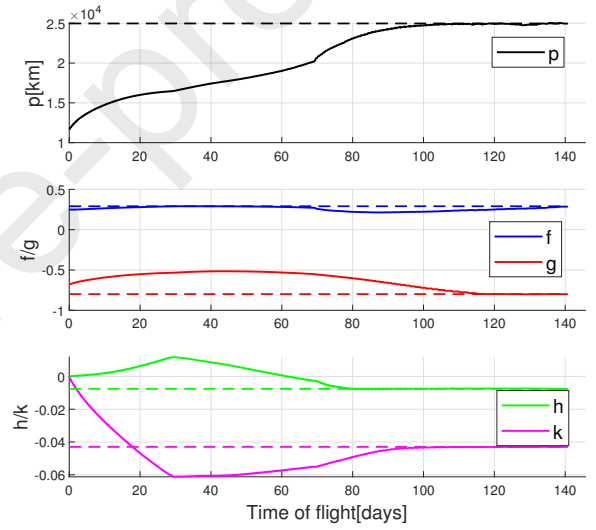
(a) Transfer trajectory without eclipse



(b) Transfer trajectory with eclipse



(c) Time history of orbital elements without eclipse



(d) Time history of orbital elements with eclipse

Fig. 17: Comparison of forward-time transfer in scenario 1 ($\sigma = 200 \text{ m}^2/\text{kg}$, $\theta_{S0}^{P1} = 65 \text{ deg}$).

Table 10: TOFs of Phase 1 in GTO-Halo transfer with eclipse.

L1/L2 halo	$\sigma [\text{m}^2/\text{kg}]$	$\theta_{S0}^{P2} [\text{deg}]$	TOF [days]
L1	100	230	115.54
	120	230	98.96
	140	230	86.08
	160	230	75.32
	180	230	67.09
	200	230	62.66
L2	100	50	166.56
	120	50	155.17
	140	50	143.17
	160	50	133.16
	180	50	119.76
	200	50	116.29

6. Conclusions

A complete design methodology for solar-sail Earth–Moon transfer trajectories, integrating a solar-sailing Q-Law with a forward–backward multiple-shooting framework, is presented in this paper. By patching the three phases, dynamically consistent trajectories from GEO or GTO to L1/L2 halo orbits are systematically constructed. Numerical simulations demonstrate that the proposed methodology reliably converges across a wide range of initial Sun phase angles and area-to-mass ratios. The results show that the transfer time is dominated by Phase 1 and decreases significantly with increasing sail acceleration, whereas Phase 3 remains nearly ballistic and insensitive to sail performance. The impact of solar eclipse in Phase 1 is found to delay the transfer for a few days compared with that without eclipse. Overall, the presented framework closes a methodological gap in solar-sail mission design by providing a unified, computationally tractable approach for Earth–Moon transfers, offering practical utility for future solar-sail exploration missions.

Appendix A. Transformation from E-M rotating frame to ECI frame

Let $\mathbf{X} \in \mathbb{R}^{6 \times 1}$ denote a state vector expressed in the Earth-centered inertial (ECI) frame and $\mathbf{X}^R \in \mathbb{R}^{6 \times 1}$ the corresponding state expressed in the Earth–Moon barycentric rotating frame of the CR3BP. The transformation from the Earth–Moon rotating frame to the ECI frame is written as

$$\mathbf{X} = \mathbf{C}_R^I \mathbf{X}_{E,d}^R \quad (\text{A.1})$$

where $\mathbf{C}_R^I \in \mathbb{R}^{6 \times 6}$ represents the transformation matrix from E-M rotating frame to ECI frame, $\mathbf{X}_{E,d}^R$ is the dimensional state vector expressed in the Earth-centered rotating frame (i.e., the barycentric rotating frame translated such that its origin coincides with the Earth's center). The nondimensional Earth-centered rotating state $\mathbf{X}_{E,nd}^R$ is obtained from the barycentric rotating state by

$$\mathbf{X}_{E,nd}^R = \mathbf{X}^R + [\mu, 0, 0, 0, 0, 0]^T \quad (\text{A.2})$$

where μ is the mass parameter of Earth–Moon CR3BP. The dimensional state $\mathbf{X}_{E,d}^R$ is then obtained by scaling $\mathbf{X}_{E,nd}^R$ with the distance and velocity unit in E-M rotating frame. Finally, the transformation matrix is

$$\mathbf{C}_R^I = \begin{bmatrix} \mathbf{C} & \mathbf{0}_3 \\ \dot{\mathbf{C}} & \mathbf{C} \end{bmatrix} \quad (\text{A.3})$$

$$\mathbf{C} = \begin{bmatrix} \hat{\mathbf{r}}_x & \hat{\mathbf{r}}_y & \hat{\mathbf{r}}_z \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} \quad (\text{A.4})$$

$$\dot{\mathbf{C}} = \begin{bmatrix} \dot{\theta}_M C_{12} & -\dot{\theta}_M C_{11} & 0 \\ \dot{\theta}_M C_{22} & -\dot{\theta}_M C_{21} & 0 \\ \dot{\theta}_M C_{32} & -\dot{\theta}_M C_{31} & 0 \end{bmatrix} \quad (\text{A.5})$$

where $\dot{\theta}_M$ is the lunar orbital angular velocity in inertial frame, $\hat{\mathbf{r}}_x, \hat{\mathbf{r}}_y, \hat{\mathbf{r}}_z$ are the instantaneous coordinates of E-M rotating frame expressed in ECI frame. Similarly, The transformation from the ECI frame to the Earth–Moon rotating frame follows the inverse procedure: 1). Dimensionalizing the ECI state using the ECI distance and velocity units; 2). Applying the inertial-to-rotating transformation matrix; 3). Non-dimensionalizing the resulting state using the CR3BP characteristic units; 4). Translating the state so that its base point is located at the barycenter of the Earth–Moon system.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: